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Psychometric properties of self-reported financial toxicity measures in cancer survivors: A systematic review

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Psychometric properties of self-reported financial toxicity measures in cancer survivors: A systematic review

Abstract

Objective: The aim of this systematic review was to summarize the psychometric properties of patient-reported outcome measures (PROMs) measuring financial toxicity (FT) in cancer survivors.

Design: Comprehensive searches were performed in seven databases. Quality appraisal, data extraction, data synthesis, and assessment of the certainty of evidence were conducted according to the guidance of the Consensus-Based Standards for the Selection of Health Measurement Instruments (COSMIN) methodology.

Results: A total of 9399 articles were identified via databases. Finally, a total of 24 articles (21 studies, 20 PROMs) were eligible for inclusion in this study. The findings highlighted that the COST had an adequate development process and showed better psychometric properties than other PROMs, especially in internal consistency, reliability, and hypothesis testing.

Conclusions: The COST could be recommended as the most suitable measures for use in research and clinical practice. The FIT and the IOC-CS financial problems domain also had adequate psychometric properties. We suggest that PROMs should

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be selected only after careful consideration of the local socioeconomic context. Future studies are warranted to develop various FT PROMs based on different social and cultural backgrounds and to clarify the theoretical grounds for assessing FT.

Keywords: cancer, financial toxicity, measurement, PROM, systematic review

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Strength and limitation of this study

- This is the first systematic review that comprehensively summarized the psychometric properties of 20 PROMs evaluating financial toxicity in cancer survivors.
- The results may provide quantitative evidence for researchers and healthcare professionals to choose PROMs measuring cancer survivors' financial toxicity in future scientific research and clinical practice.
- This review only included studies that aimed to evaluate the measurement properties of financial toxicity PROMs.

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Introduction

The rising cost associated with advancements in cancer treatment and lengthening of cancer survivorship poses a significant challenge to survivors, caregivers, and public healthcare systems.[1,2] Total global spending on cancer medications grows at a compound annual growth rate of 6.5% from US$ 96 billion in 2013 to US$ 173 billion in 2020, which is nearly twice the rate of global gross domestic product (GDP) growth.[3-5] The majority of cancer survivors in middle-income and low-income countries/regions depend on out-of-pocket payments, which may lead to global inequalities in healthcare expenditures and financial insecurity for vulnerable groups.[6,7]

The term "financial toxicity (FT)" has been described as the economic effect of cancer treatment in the age of precision medicine.[2,8,9] Witte and colleagues described FT as "the patient-reported outcome (PRO) of perceived subjective financial distress resulting from objective financial burden".[10] This concept covers both the objective financial burden and the subjective financial distress that cancer survivors face as a result of high out-of-pocket medical expenses. Regarding the terminology, "financial toxicity", "financial burden" and "financial distress" are often used interchangeably in research and share a similar definition.[10,11] In this review, the authors agreed to consistently use the term "financial toxicity." Financial toxicity is usually measured by patient-reported outcome measures (PROMs); choosing a PROM with high validity and reliability is a prerequisite for robust results.

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There are a few cancer-specific and generic FT PROMs that have been reported and used in different contexts. As one of the recent cancer-specific FT PROMs, the Comprehensive Score for Financial Toxicity (COST) is the most commonly used measure for assessing FT.[12] In addition to COST, other cancer-specific measures have been widely used, including the Breast Cancer Finances Survey Inventory (BCFS),[13] Socioeconomic Wellbeing Scale (SWBS),[14] and InCharge Financial Distress/Financial Wellbeing Scale (InCharge).[15] Additionally, validated subscales, such as the Social Difficulties Inventory Cancer Care Outcomes (SDI) and the Cancer Care Outcomes Research and Surveillance Consortium patient survey (CanCORS), were also used to evaluate FT.[16,17] However, existing PROMs vary significantly in their state of development and degree of validation, and many PROMs have not been psychometrically tested.

A preliminary literature search was conducted in PubMed, PsycINFO (EBSCO), Cochrane Library (Wiley) and JBI (Ovid), which revealed that there exist some reviews regarding measures of FT. Witte and colleagues summarized the content of 352 items from 34 studies measuring FT in cancer survivors.[10] However, this review did not report the psychometric properties of the included PROMs, and most of the included PROMs were not validated through a scientific process, which made it difficult for readers to choose the best measure from existing PROMs to evaluate the level of FT. Salman and colleagues conducted a COSMIN systematic review and

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found 8 PROMs and 2 caregiver-reported measures for assessing financial burden in adolescents and young adults.[18] However, this review focused only on PROMs assessing FT in adolescents and young adults with cancer. The psychological properties of FT measures in adult cancer survivors are still unknown.

The reproducibility, reliability, and accuracy of PROMs are the fundamental premise for achieving robust results.[22,23] Therefore, it is necessary to summarize the psychometric properties of existing PROMs for future research. However, this information is still lacking. The aim of this systematic review was to summarize the psychometric properties of PROMs for measuring FT in cancer survivors. The review was conducted according to the guidance of the Consensus-Based Standards for the Selection of Health Measurement Instruments (COSMIN) methodology and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[19,20] The protocol of this review was published in *BMJ OPEN* in 2020.[21] The registration number of protocol in PROSPERO was CRD42021254721.

Methods

Search strategy

First, we conducted a limited search via PubMed to capture keywords from which to develop search strategies for each database. Subsequently, all identified search strategies across databases were performed in PubMed, MEDLINE (Ovid), Embase (Ovid), CINAHL (EBSCO), Web of Science, ProQuest Dissertations and

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Theses, and Cochrane Library (Wiley). In PubMed/Medline, we searched papers in English using MeSH terms ([cancer OR neoplasms] AND ["cancer survivors" OR patient OR survivors] AND "cost of illness") combined with (cancer OR [patient* OR survivor*] AND [cost OR bill* OR expense OR productivity loss OR "out-of-pocket" OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden"]). The COSMIN measurement properties filter and exclusion filter were also utilized in the search box. The search strategies for each database are presented in Appendix 1. Finally, the references of all included studies were manually reviewed to supplement the database search.

Inclusion and exclusion criteria

The inclusion criteria were as follows: 1) studies that reported any PROM for measuring FT in cancer survivors who were ≥18 yrs old; 2) studies that evaluated at least one measurement property; and 3) studies published in English. The exclusion criteria were as follows: 1) studies that were not validation studies and used a PROM only as an outcome measurement; 2) studies that used a PROM as a comparator for another instrument; 3) studies that did not provide empirical data; additionally, 4) if a measure was a quality of life PROM and had a domain that assessed FT, we included only the original version of the PROM.

Study screening and selection

We imported all identified citations by search strategies into Endnote X8

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(Clarivate Analytics, PA, USA). After duplicates were removed, two reviewers (ZZ & WX) independently screened all titles, abstracts, and full texts (ZZ & WX) based on the established inclusion and exclusion criteria. Any disagreements were resolved by a third reviewer (YH).

Quality appraisal

Two reviewers (WH & YS) assessed the methodological quality of the PROM of the included studies by using the COSMIN Risk of Bias Checklist. The checklist consisted of 10 domains (38 items), including PROM development, content validity, structural validity, internal consistency, cross-cultural validity, reliability, measurement error, criterion validity, hypothesis testing, and responsiveness. Each measurement property was rated as “very good”, “adequate”, “doubtful” or “inadequate quality”. The methodological quality of a single study is rated based on the worst score counts method. Any discrepancies were resolved by a third reviewer (ZZ).

Data extraction

Two reviewers (ZZ & WX) independently extracted data from the included papers, including authors, year of publication, PROM, country/language, study design, target population, sample size, domains, number of items, total score range, and main findings. The main findings regarding psychometric properties including content validity, structural validity, internal consistency, cross-cultural validity,

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reliability, measurement error, criterion validity, hypothesis testing, and responsiveness, were also extracted. Any discrepancies were resolved through discussion between the two reviewers.

Data synthesis

We used the COSMIN criteria to summarize and evaluate the psychometric properties of each study regarding structural validity, internal consistency, reliability, measurement error, hypothesis testing for construct validity, cross-cultural validity/measurement invariance, criterion validity, and responsiveness. Each measurement property from each study was rated as sufficient (+), insufficient (−), or indeterminate (?). If the ratings of one psychometric property per study were all sufficient or insufficient, the results were pooled, and the overall rating was rated as sufficient or insufficient. If the ratings were inconsistent, explanations of inconsistency were explored (e.g., different languages). If the explanation was reasonable, we provided ratings in subgroups. If the explanation was not reasonable, the overall rating of this measurement property was rated as inconsistent (±). If there was no information supporting the rating, the overall rating was indeterminate (?).

Assessing certainty of evidence

We used a modified Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system to assess the certainty of evidence. Each piece of evidence was graded for risk of bias, inconsistency, imprecision, and indirectness.

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Four reviewers (ZZ, WJ, HW, and YS) independently assessed the grade. Any discrepancies were resolved by discussion.

Results

Literature search

Figure 1 shows the process of literature screening and selection. A total of 9399 articles were identified via databases. Six articles were found by additional supplementary searches. After duplications were removed, a total of 9405 articles were retained, 9348 articles were deleted after reading the title and abstract, and 58 were deleted after full-text reading. Finally, a total of 24 articles (21 studies, 20 PROMs) were eligible for inclusion in this study.[12,14,16,17,22-41]

Study description

Table 1 shows the characteristics of the included studies. All included studies were published from 1988 to 2021. Seven studies were conducted in the United States,[12,17,24,27,37,41] four in the United Kingdom,[16,29,35,38] two in Canada,[31,36] and two in China (mainland and Hong Kong).[25,39] One study was conducted in 12 counties in Europe and North America.[22,23] A total of 12412 participants were included, ranging from 7[36] to 5901[41] per study. The majority of studies assessed FT in multiple types of cancer. Only three studies focused on a single type of cancer, namely, lung, colorectal, or head and neck cancer.[22,23,31,37]

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Among the 20 PROMs, 9 were FT-related domains of quality of life PROMs, and 11 were independent PROMs focusing on FT. All PROMs were validated in cancer survivors. Thirteen PROMs were in English,[12,14,16,17,24,26-31,35,37,38,40,41] and two were in Chinese.[25,39] Other languages included French,[36] Portuguese,[32] Italian,[33] and Persian.[34] The number of items evaluating FT ranged from 1[22,23] to 23.[36] Cancer Care Outcomes Research and Surveillance (CanCORS) did not report the items or total score range of the FT-related domain.[17] The French version of the Patient Self-Administered Financial Effects questionnaire (P-SAFE) did not report the total score range of the whole PROM.[36]

Quality assessment

Methodological quality assessment. Table 2 shows the methodological quality of the 21 included studies by using the COSMIN checklist. In the PROM development domain, only three studies were rated as doubt, and the others were rated as inadequate.[12,25,27,29] Two studies reported adequate information in testing the relevance, comprehensiveness, and comprehensibility of PROMs.[12,27,29] Among all studies, the most reported domain was internal consistency, except three studies.[17,22,23,36] Limited information could be retrieved on cross-cultural validity (three studies),[31,32,36] criterion validity (six studies),[16,24,33,35,38,40] reliability (nine studies),[12,16,25,27,28,33,35,38,39,40] and responsiveness (two studies).[31,39] No data were identified on measurement error.

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Measurement property assessment. Table 3 shows the quality of the psychometric properties retrieved from 20 PROMs. Only the Persian version of the COST-v2 was rated as "+" in structural validity.[34] There were 16 PROMs rated as "+" in internal consistency.[12,14,16,24,25,27-29,31,32,34,35,37-39,41] Six PROMs were rated as "+" in reliability.[12,25,27-29,31,35] Seven PROMs were rated as "+" in hypothesis testing.[12,14,24,25,27-31] Limited information was retrieved on cross-cultural validity (two PROMs),[32,36] criterion validity (six PROMs),[16,24,33,35,38,40] and responsiveness (two PROMs).[39,31] No PROMs reported data on measurement error.

Certainty of evidence

Table 4 shows the certainty of evidence for each measurement property. Among all included PROMs, COST showed the best psychometric properties compared to other measures. COST and its seven versions were rated as high evidence in structural validity, internal consistency, hypothesis testing, and criterion validity.[12,25-28,32,33,34,39] Financial Index of Toxicity (FIT) and Impact of Cancer – Childhood Survivors IOC-CS financial problems domain reported data on five properties and were rated as very low evidence to high evidence.[31,40] The European Organisation for Research and Treatment of Cancer (EORTC) Core Quality of Life Questionnaire (QLQ-C30) financial impact domain and CanCORS insurance coverage and income domain did not provide any qualified data on all properties.[17,22,23]

Discussion

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This systematic review identified 20 PROMs and domains of PROMs evaluating FT in cancer survivors, including the COST (original, Brazilian, India, Italian, Persian, Simplified Chinese, Traditional Chinese version), FIT, personal financial burden (PFB), P-SAFE, SWBS, EORTC QLQ-C30 financial impact domain, Quality of Life in Adult Cancer Survivors (QLACS) financial problems domain, Chronic Cancer Experiences Questionnaire (CCEQ) financial advice domain, CanCORS insurance coverage and income domain, Patient Roles and Responsibilities Scale (PRRS) financial well-being domain, SDI-21 providing for the family domain, SDI-16 money matters domain, IOC-CS financial problems domain, and Cancer Problems in Living Scale (CPILS) employment/financial domain. Overall, the COST had a complete development process compared to other PROMs and showed the best psychometric properties, especially in terms of internal consistency, reliability, and hypothesis testing. To the best of our knowledge, this is the first systematic review that has summarized the psychometric properties of FT PROMs in cancer survivors and reported the certainty of evidence for each property of PROMs. The results may provide quantitative evidence for researchers and healthcare professionals to choose PROMs measuring cancer survivors' FT in future scientific research and clinical practice.

The results highlighted that the COST (of which we studied both version 1 and version 2) had better psychometric properties than other specific and generic PROMs in terms of internal consistency, reliability, and hypothesis testing. The COST could

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be recommended as a suitable PROM in research and clinical practice. Other systematic reviews have also suggested that the COST is a promising measure from a content perspective[10,11]. From a psychometric standpoint, there are a few issues that one must face when evaluating financial toxicity in cancer survivors using the COST. First, caution should be taken when using the COST in different socioeconomic conditions outside the United States. In some countries in Europe or Asia, the majority of medical expenses are covered by social health insurance, and direct out-of-pocket payments are replaced by prepayment from health insurance contributions.[42,43] In addition, social security systems can also benefit cancer survivors who are not able to work.[44] These two socioeconomic factors may affect cancer survivors' understanding regarding some items related to medical spends and indirect cost. However, few COST validation studies have considered socioeconomic issues, adapted the measure in a local context, or provided data on cross-cultural validity. It is recommended that future COST validation studies recruit cancer survivors across multiple social and cultural backgrounds to assess cross-cultural measurement invariance.

Second, the original construct and item generation for the COST were based on a literature search; thus, the theoretical grounds for the measure are unclear, and the instrument may not capture detailed information related to the construct. Theoretical frameworks and conceptual models are crucial for self-reported measures to capture subtle changes in constructs.[45] Although FT is a relatively new concept, certain

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models can guide item generation in the development of future FT PROMs. Tucker-Seeley and colleagues developed a conceptual model of FT and emphasized three components of financial burden, namely, the material, psychosocial, and behavioral domains.[46] Head developed SWBS based on James Coleman's Theory of Social Class; this scale contains 17 items across 3 domains: including human capital, material capital, and social capital.[14,30,47] Witte and colleagues' systematic review analyzed 352 different questions regarding financial spending and found six domains (financial spending, financial resources, psychosocial affect, support seeking, coping care, and coping lifestyle) that can represent reactions to subjective financial distress.[10] Other theories and models, including the Wreckers theory of financial distress, ecological theory, and the functionalist tradition, have also been widely used in cancer survivors.[48-50] With the increasing number of theoretical studies related to FT, the theoretical grounds for future PROMs need to be clarified.

In addition to the COST, two other PROMs, namely the FIT and the IOC-CS financial problems domain, also provided adequate data on psychometric properties. FIT is relatively new and has fewer items than other included measures. This measure was developed by Hueniken and colleagues and has been validated only in survivors with head and neck cancer.[31] Head and neck cancer, especially laryngeal and hypopharyngeal cancer, has particularly large impacts on survivors' daily function (e.g., speech and eating) after treatment and affects survivors' ability to return to work.[51,52] Only 32% to 59% of head and neck cancer survivors return to work after

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treatments.[53] This form of cancer also has short- and long-term financial consequences for caregivers and their families.[54] Therefore, future studies should be aware that the FIT may not be directly applicable to other cancer populations.

Regarding PROM development, we found that all original versions of FT PROMs were established in developed countries. Previous studies have reported that FT is closely related to broad social determinants of economic circumstances, such as healthcare policy and the level of regional economic development.[55,56] Additionally, cultural factors (e.g., a cultural emphasis on saving and a cultural imperative to have a large family) also affect cancer survivors' perceived financial security and economic burden.[57] Therefore, further studies are warranted to develop various FT PROMs based on different social and cultural backgrounds.

Limitations

We acknowledge that there are some limitations to this study. First, this review included only studies that aimed to evaluate the measurement properties of FT PROMs. Many studies that aimed to explore the level of FT in cancer survivors also reported the reliability and validity of PROMs. Therefore, the PROMs we summarized in this systematic review had higher psychometric quality than other measures that we did not list in this review. Second, we included only studies published in English. Therefore, studies published in other languages were not included, which may affect the conclusion of this review. Third, we included only the

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original version of the FT domain from PROMs assessing quality of life in cancer survivors, such as the EORTC QLQ-C30 and the QLACS. Over 20 language versions of these PROMs do not provide sufficient details on the FT domain individually.

Conclusion

This systematic review summarized the psychometric properties of 20 PROMs evaluating FT in cancer survivors. The findings highlighted that the COST had an adequate PROM development process and showed the best psychometric properties among all examined PROMs, especially in internal consistency, reliability, and hypothesis testing; thus, we recommend the COST as the most suitable measure for use in research and clinical practice. The FIT and the IOC-CS financial problems domain also had adequate psychometric properties. We suggest that PROMs should be selected only after careful consideration of the local socioeconomic context. Future studies are warranted to develop various FT PROMs based on different social and cultural backgrounds and a clear theoretical basis for assessing FT.

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Contributors

ZZ and WX designed the systematic review, conducted data searching, extraction and analysis, assessing the certainty of evidence, and wrote the draft of the manuscript. HW and YS conducted quality appraisal and assessing the certainty of evidence. WS, LL, JP, and YH provided critical comments. All authors approved the

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Competing interests

None.

Patient and public involvement

Patients and the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Patient consent for publication

Not applicable.

Ethics approval

This systematic review did not require ethical approval.

Data available statement

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No additional data are available.

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Table 1 Overview of the included studies

Author (year), country	PROM	Country	Language(s) of PROM	Study design	Target population	Sample size	Measurement domain	Number of items	Total score range
Aaronson et al., 1988 Aaronson et al., 1993	EORTC QLQ-C30 Financial impact domain	12 countries in Europe and North America	9 languages	Development study and validation study	Age: 63±10 yrs Male: 76% Cancer type: Lung cancer	305	1 domain regarding perceived financial impact	1	1-4
Avis et al., 2005	QLACS financial problems domain	US	English	Development and validation study	Age: 71.4±11.5 yrs Male: 42% Cancer type: Breast, bladder, head and neck, gynecologic, prostate, colorectal cancer	242	1 domain regarding financial problems	4	4-28
Chan et al., 2021	COST-v2, *Traditional Chinese version*	Hong Kong	Traditional Chinese	Validation study	Age: 59.9±11.1 yrs Male: 35.3% Cancer type: Breast, gynecological, head and neck, gastric and colorectal, genitourinary, lung, hematological, skin, bone and soft tissue, brain and central nervous system cancer and others	640	No subdomain	12	0-44
Dar et al., 2021	COST-v1, *India version*	India	Hindi or English	Validation study	Age: 49.5±16.8 yrs Male: 82.8% Cancer type: Tongue, gingival buccal sulcus, buccal mucosa, supraglottic larynx, hypopharynx, parotid and others	29	No subdomain	11	0-44
de Alcantara Nogueira et al., 2020	COST-v1, *Brazilian version*	Brazil	Brazilian Portuguese	Validation study	Mean age: 56 yrs Male: 40.5% Cancer type: Not specific	126	No subdomain	11	0-44

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De Souza et al., 2014 De Souza et al., 2017	COST-v1	US	English	Development study and validation study	Age: 58.4±11.5 yrs Male: 41.6% Cancer type: Not specified (diagnosis of AJCC stage IV cancer)	233	No subdomain	11	0-44
Durber et al., 2021	COST-v1	Australia	English	Validation study	Age: ≤50 yrs 23%; 51-64 yrs 30%;≥65 yrs 48% Male: 46% Cancer type: Thoracic, breast, sarcoma, skin, central nervous system, gynecological, head and neck, colorectal, upper gastrointestinal, urological, and miscellaneous cancer	257	No subdomain	11	0-44
Harley et al., 2019	CCEQ financial advice domain	UK	English	Development and validation study	*Pilot study:* Age: 65 (41-90) yrs Male: 48.5% Cancer type: Breast, colorectal/gastrointestinal, gynecological, prostate, and renal cancer *Final study:* Age: 67 (41-88) yrs Male: 50.0% Cancer type: Breast, colorectal/gastrointestinal, gynecological, prostate, and renal cancer	103 for pilot study 313 for final study	1 domain regarding financial advice	5	5-25
Head, 2008 Head & Faul, 2008	SWBS	US	English	Development and validation study	Age: 59.6±12.7 yrs Male: 35.7% Cancer type: Breast, melanoma, head and neck, prostate, rectum/anus, colon, endometrium, lung/tracheal/bronchus, and	266	Two domains: material and social capital.	17	0-68

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					non-Hodgkin's lymphoma				
Hueniken et al., 2020	FIT	Canada	English	Development and validation study	Age: 61.6 (25.5-88.5) yrs Male: 77.2% Cancer type: Oropharynx, oral cavity, larynx, nasopharynx, hypopharynx cancers, and others	430	No subdomain	9	0-100
Malin et al., 2006	CanCORS insurance coverage and income domain	US	English	Development and validation study	Age: 66±11 yrs Male: 51% Cancer type: Lung and colorectal cancer	71	1 domain regarding insurance coverage and income	NR	NR
Ripamonti et al., 2020	COST-v2, *Italian version*	Italy	Italian	Validation study	Age: 61.5±12.7 yrs Male: 52.5% Cancer type: Breast, lung, colon, gastric, hepatocellular, endometrial, prostate, sarcoma, bladder, head and neck, Hodgkin lymphoma, non-Hodgkin lymphoma, leukemia, myeloma, and others	118	No subdomain	11	0-44
Sharif et al., 2020	COST-v2, *Persian version*	Iran	Persian	Validation study	Age: 50.0±14.3 yrs Male: 51.0% Cancer type: Not specific	398	No subdomain	11	0-44
Shilling et al., 2018	PRRS financial wellbeing domain	UK	English	Development and validation study	Age: ≤50 yrs 25%; 51-65 yrs 41%; ≥66 yrs 34% Male: 23% Cancer type: breast, gynecological, lung, and melanoma cancers	135	1 domain regarding financial wellbeing	6	0-24
Tremblay et al., 2020	P-SAFE, *French Version*	Canada	French	Cross-adaption study	Age: 50-59 yrs 57%; 60-69 yrs 29%; ≥70 yrs 34% Male: 14% Cancer type: colorectal, lung, breast, and prostate cancer.	7	NR	23	NR
Veenstra et al., 2014	PFB	US	English	Validation study	Age: <50 yrs 17%; 50-64 yrs 37%; 65-74 yrs 23%; >75 yrs 24%	956	No subdomain	7	0-7

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					Male: 53% Cancer type: Stage III colorectal cancer				
Wright et al., 2005	SDI-21 providing for the family domain	UK	English	Development and validation study	Age: 53.8±14.1 yrs Male: Not specific Cancer type: brain, lung cancers and others	271	1 domain regarding providing for the family	5	0-20
Wright et al., 2011	SDI-16 money matters domain	UK	English	Development and validation study	Age: 56 (18-88) yrs for men; 56 (21-88) yrs for women Male: 48% Cancer type: breast, gastrointestinal, hematology, gynecological, germ cell, head and neck, lung, genitourinary, and others	652	1 domain regarding money matters	5	0-20
Yu et al., 2021	COST-v1, *Simplified Chinese version*	Mainland China	Chinese	Validation study	Age: 57.0±9.2 yrs Male: 45.7% Cancer type: lung, stomach, colorectal, and breast cancer	440	No subdomain	11	0-44
Zebrack et al., 2010	IOC-CS financial problems domain	US	English	Validation study	Age: 26.7±5.3 yrs Male: 48.0% Cancer type: hematological, brain, and solid tumors/soft tissue tumors	519	1 domain regarding financial problems	3	1-15
Zhao et al., 2009	CPILS employment/financial domain	US	English	Validation study	Age: ≤55 yrs 48.8%; >55 51.2% Male: 41.6% Cancer type: breast, prostate, colorectal, bladder, uterine, kidney, lung, and ovarian cancer; melanoma of skin; non-Hodgkin lymphoma,	5901	1 domain regarding employment/finances	6	0-12

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Table 2 Methodological quality assessment of the measures

Author (year)	PROM	Measurement property: methodological quality by study									
		PROM development	Content validity	Structural validity	Internal consistency	Cross-cultural validity	Reliability	Measurement error	Criterion validity	Hypothesis testing	Responsiveness
Aaronson et al., 1988 Aaronson et al., 1993	EORTC QLQ-C30 Financial impact domain	Inadequate	R: NR C1: NR C2: NR	NR	NA	NR	NR	NR	NR	NR	NR
Avis et al., 2005	QLACS financial problems domain	Inadequate	R: NR C1: NR C2: NR	Adequate	Very good	NR	NR	NR	Adequate	Doubtful	NR
Chan et al., 2021	COST-v2, *Traditional Chinese version*	Doubtful	R: Doubtful C1: NR C2: Doubtful	Very good	Very good	NR	Doubtful	NR	NR	Very good	NR
Dar et al., 2021	COST-v1, *India version*	Inadequate	R: NR C1: NR C2: NR	Inadequate	Very good	NR	NR	NR	NR	NR	NR
de Alcantara Nogueira et al., 2020	COST-v1, *Brazilian version*	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	Inadequate	NR	NR	NR	NR	NR
De Souza et al., 2014 De Souza et al., 2017	COST-v1	Doubtful	R: Adequate C1: Adequate C2: Adequate	NR	Very good	NR	Adequate	NR	NR	Very good	NR
Durber et al., 2021	COST-v1, *Australia*	Inadequate	R: NR C1: NR	NR	Very good	NR	Adequate	NR	NR	Very good	NR

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	version		C2: NR								
Harley et al., 2019	CCEQ financial advice domain	Doubtful	R: Adequate C1: Adequate C2: Adequate	Adequate	Very good	NR	NR	NR	NR	Very good	NR
Head, 2008 Head & Faul, 2008	SWBS	Inadequate	R: NR C1:Doubt C2: NR	Very good	Very good	NR	NR	NR	NR	Very good	NR
Hueniken et al., 2020	FIT	Inadequate	R: NR C1: NR C2: NR	Adequate	Very good	Doubtful	NR	NR	NR	Very good	Very good
Malin et al., 2006	CanCORS insurance coverage and income domain	Inadequate	R: NR C1: NR C2: NR	NR	NR	NR	NR	NR	NR	NR	NR
Ripamonti et al., 2020	COST-v2, *Italian version*	Inadequate	R: NR C1: NR C2: NR	Inadequate	Very good	NR	Inadequate	NR	Very good	Very good	NR
Sharif et al., 2020	COST-v2, *Persian version*	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	NR	NR	NR	NR	Inadequate	NR
Shilling et al., 2018	PRRS financial wellbeing domain	Inadequate	R: NR C1: NR C2: NR	Adequate	Very good	NR	Inadequate	NR	Inadequate	NR	NR
Tremblay et al., 2020	P-SAFE, *French Version*	Inadequate	R: NR C1: NR C2: NR	NR	NR	Inadequate	NR	NR	NR	NR	NR
Veenstra et al., 2014	PFB	Inadequate	R: NR C1: NR C2: NR	Very good	Inadequate	NR	NR	NR	NR	NR	NR
Wright et al., 2005	SDI-21 providing for	Inadequate	R: Doubt C1: NR	Very good	Very good	NR	Inadequate	NR	Inadequate	NR	NR

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	the family domain		C2: NR								
Wright et al., 2011	SDI-16 money matters domain	Inadequate	R: Doubt C1: NR C2: NR	Very good	Very good	NR	Inadequate	NR	Inadequate	NR	NR
Yu et al., 2021	COST-v1, *Simplified Chinese version*	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	NR	Very good	NR	NR	Doubt	Doubt
Zebrack et al., 2010	IOC-CS financial problems domain	Inadequate	R: NR C1: NR C2: Doubt	Adequate	Very good	NR	Adequate	NR	Inadequate	Very good	NR
Zhao et al., 2009	CPILS employment/financial domain	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	NR	NR	NR	NR	Very good	NR

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569 **Table 3** Rating of measurement properties

PROM	Author (year)	Structural validity	Internal consistency	Reliability	Measurement error	Hypothesis testing	Cross-cultural validity	Criterion validity	Responsiveness
CanCORS insurance coverage and income domain	Malin et al., 2006	NR	NR	NR	NR	NR	NR	NR	NR
CCEQ financial advice domain	Harley et al., 2019	NR	+	NR	NR	+	NR	NR	NR
COST-v1	De Souza et al., 2014; De Souza et al., 2017	NR	+	+	NR	+	NR	NR	NR
COST-v1, *Australia version*	Durber et al., 2021	NR	+	+	NR	+	NR	NR	NR
COST-v1, *Brazilian version*	de Alcantara Nogueira et al., 2020	-	+	NR	NR	NR	+	NR	NR
COST-v1, *India version*	Dar et al., 2021	-	?	NR	NR	NR	NR	NR	NR
COST-v1 *Simplified Chinese version*	Yu et al., 2021	-	+	+	NR	?	NR	NR	?
COST-v2, *Italian version*	Ripamonti et al., 2020	?	?	-	NR	?	NR	?	NR
COST-v2, *Persian version*	Sharif et al., 2020	+	+	NR	NR	?	NR	NR	NR
COST-v2, *Traditional Chinese version*	Chan et al., 2021	-	+	+	NR	+	NR	NR	NR
CPILS employment/financial domain	Zhao et al., 2009	?	+	NR	NR	?	NR	NR	NR
EORTC QLQ-C30 financial impact domain	Aaronson et al., 1988; Aaronson et al., 1993	NR	NA	NR	NR	NR	NR	NR	NR
FIT	Hueniken et al., 2020	?	+	+	NR	+	NR	NR	+

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IOC-CS financial problems domain	Zebrack et al., 2010	?	-	-	NR	?	NR	-	NR
PFB	Veenstra et al., 2014	?	+	NR	NR	NR	NR	NR	NR
PRRS financial wellbeing domain	Shilling et al., 2018	?	+	+	NR	NR	NR	+	NR
P-SAFE, *French Version*	Tremblay et al., 2020	NR	NR	NR	NR	NR	+	NR	NR
QLACS financial problems domain	Avis et al., 2005	-	+	NR	NR	+	NR	-	NR
SDI-16 money matters domain	Wright et al., 2011	?	+	-	NR	NR	NR	?	NR
SDI-21 providing for the family domain	Wright et al., 2005	?	+	-	NR	NR	NR	?	NR
SWBS	Head, 2008; Head & Faul, 2008	?	+	NR	NR	+	NR	NR	NR

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571 **Table 4** Certainty of evidence of measurement properties

PROM	Author (year)	Structural validity	Internal consistency	Reliability	Measurement error	Hypothesis testing	Cross-cultural validity	Criterion validity	Responsiveness
CanCORS insurance coverage and income domain	Malin et al., 2006	-	-	-	-	-	-	-	-
CCEQ financial advice domain	Harley et al., 2019	Moderate	High	-	-	High	-	-	-
COST-v1	De Souza et al., 2014; De Souza et al., 2017	-	High	Moderate	-	High	-	-	-
COST-v1, *Australia version*	Durber et al., 2021	-	High	Moderate	-	High	-	-	-
COST-v1, *Brazilian version*	de Alcantara Nogueira et al., 2020	High	High	-	-	-	Very low	-	-
COST-v1, *India version*	Dar et al., 2021	Very Low	Low	-	-	-	-	-	-
COST-v1, *Simplified Chinese version*	Yu et al., 2021	High	High	High	-	Low	-	-	Low
COST-v2, *Italian version*	Ripamonti et al., 2020	Very low	High	Very low	-	High	-	High	-
COST-v2, *Persian version*	Sharif et al., 2020	High	High	-	-	Very low	-	-	-
COST-v2, *Traditional Chinese version*	Chan et al., 2021	High	High	Low	-	High	-	-	-
CPILS employment/financial domain	Zhao et al., 2009	High	High	-	-	High	-	-	-
EORTC QLQ-C30 financial impact domain	Aaronson et al., 1988; Aaronson et al., 1993	-	-	-	-	-	-	-	-
FIT	Hueniken et al., 2020	Low	Moderate	Very Low	-	Moderate	-	-	Moderate

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IOC-CS financial problems domain	Zebrack et al., 2010	Moderate	High	Moderate	-	High	-	Very low	-
PFB	Veenstra et al., 2014	Moderate	Very low	-	-	-	-	-	-
PRRS financial wellbeing domain	Shilling et al., 2018	Moderate	High	Very low	-	-	-	Very low	-
P-SAFE, *French Version*	Tremblay et al., 2020	-	-	-	-	-	Very low	-	-
QLACS financial problems domain	Avis et al., 2005	Moderate	High	-	-	Low	-	Moderate	-
SDI-16 money matters domain	Wright et al., 2011	High	High	Very low	-	-	-	Very low	-
SDI-21 providing for the family domain	Wright et al., 2005	High	High	Very low	-	-	-	Very low	-
SWBS	Head, 2008; Head & Faul, 2008	High	High	-	-	High	-	-	-

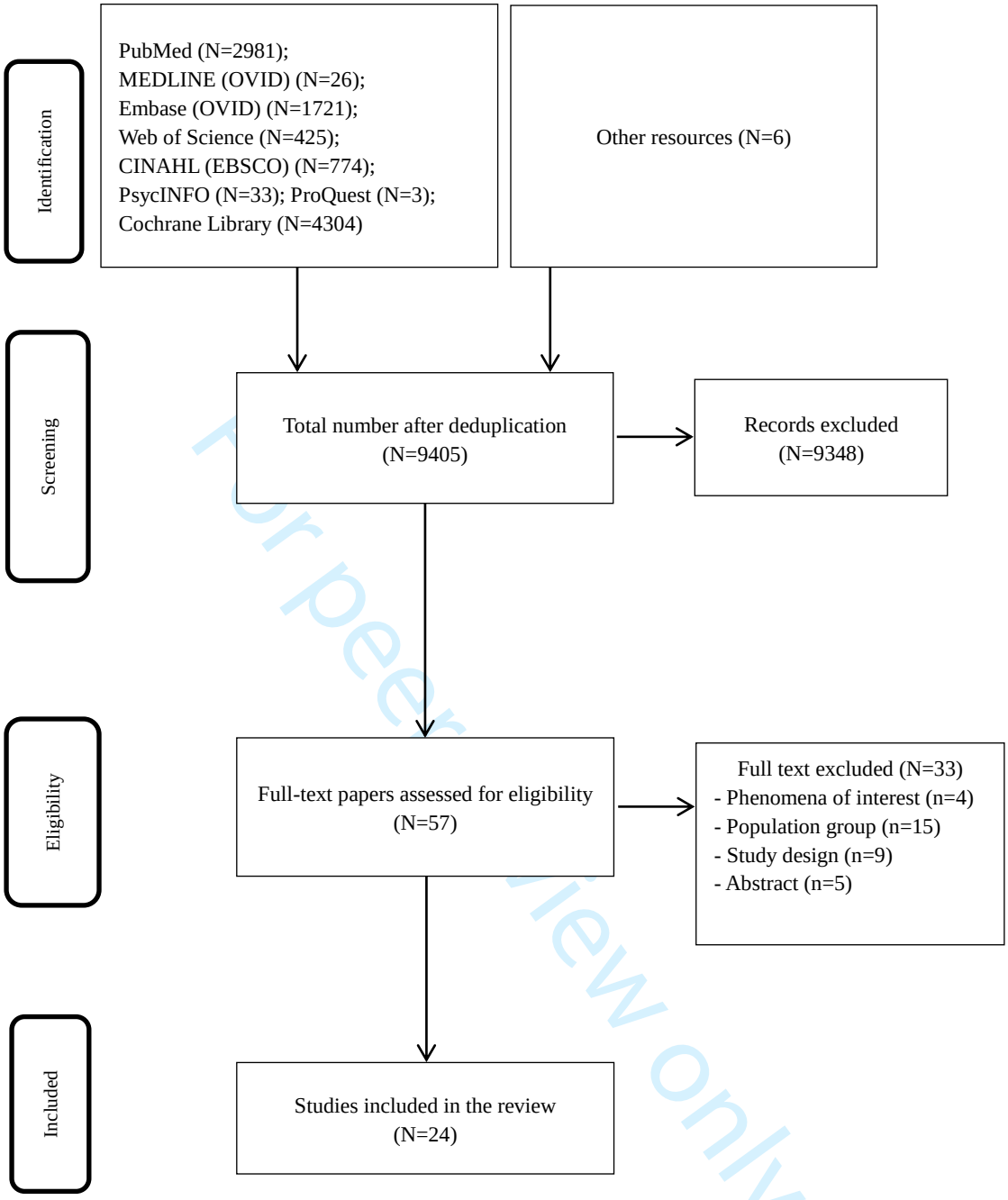
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573 **Figure 1** PRISMA Flow Chat of Selection Process

574 **Appendix 1** Search Strategy and Results

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Appendix 1 Search strategy and results

Search strategy for PubMed

Search time: 2021-5-3 14:32 (UTC+8)

Search	Query	Items found
#1	Cancer[Title/Abstract] OR neoplasms[MeSH]	3971704
#2	Patient?[Title/Abstract] OR survivor?[Title/Abstract] OR patients[MeSH] OR "cancer survivors"[MeSH] OR survivors[MeSH]	2564363
#3	Cost[Title/Abstract] OR bill?[Title/Abstract] OR expense[Title/Abstract] OR "productivity loss"[Title/Abstract] OR "out-of-pocket"[Title/Abstract] OR "economic burden"[Title/Abstract] OR "financial toxicity"[Title/Abstract] OR "financial hardship"[Title/Abstract] OR "financial burden"[Title/Abstract] OR "financial effect"[Title/Abstract] OR "financial stress"[Title/Abstract] OR "economic burden"[Title/Abstract] OR "economic hardship"[Title/Abstract] OR "co-payment"[Title/Abstract] OR "cost of illness"[MeSH]	530485
#4	Scale?[Title/Abstract] OR "patient reported outcome measur*"[Title/Abstract] OR PROM? [Title/Abstract] OR measure* [Title/Abstract] OR "Patient Reported Outcome Measures*"[MeSH] OR "Surveys and Questionnaires"[MeSH]	4782348
#5	(instrumentation[sh] OR methods[sh] OR "Validation Studies"[pt] OR "Comparative Study"[pt] OR "psychometrics"[MeSH] OR psychometr*[tiab] OR clinimetr*[tw] OR clinometr*[tw] OR "outcome assessment (health care)"[MeSH] OR "outcome assessment"[tiab] OR "outcome measure*"[tw] OR "observer variation"[MeSH] OR "observer variation"[tiab] OR "Health Status Indicators"[Mesh] OR "reproducibility of results"[MeSH] OR reproducib*[tiab] OR "discriminant analysis"[MeSH] OR reliab*[tiab] OR unreliab*[tiab] OR valid*[tiab] OR "coefficient of variation"[tiab] OR coefficient[tiab] OR homogeneity[tiab] OR homogeneous[tiab] OR "internal consistency"[tiab] OR (cronbach*[tiab] AND (alpha[tiab] OR alphas[tiab])) OR (item[tiab] AND (correlation*[tiab] OR selection*[tiab] OR reduction*[tiab])) OR agreement[tw] OR precision[tw] OR imprecision[tw] OR "precise values"[tw] OR test-retest[tiab] OR (test[tiab] AND retest[tiab]) OR (reliab*[tiab] AND (test[tiab] OR retest[tiab])) OR stability[tiab] OR interrater[tiab] OR inter-rater[tiab] OR intrarater[tiab] OR intra-rater[tiab] OR intertester[tiab] OR inter-tester[tiab] OR intratester[tiab] OR intra-tester[tiab] OR interobserver[tiab] OR inter-observer[tiab] OR intraobserver[tiab] OR intra-observer[tiab] OR intertechnician[tiab] OR inter-technician[tiab] OR intratechnician[tiab] OR intra-technician[tiab] OR interexaminer[tiab] OR inter-examiner[tiab] OR intraexaminer[tiab] OR intra-examiner[tiab] OR interassay[tiab] OR inter-assay[tiab] OR intraassay[tiab] OR intra-assay[tiab] OR interindividual[tiab] OR inter-individual[tiab] OR intraindividual[tiab] OR intra-individual[tiab] OR interparticipant[tiab] OR inter-participant[tiab] OR intraparticipant[tiab] OR intra-participant[tiab] OR kappa[tiab] OR kappa's[tiab] OR kappas[tiab] OR repeatab*[tw] OR ((replicab*[tw] OR repeated[tw]) AND (measure[tw] OR measures[tw] OR findings[tw] OR result[tw] OR results[tw] OR test[tw] OR tests[tw])) OR generaliza*[tiab] OR generalisa*[tiab] OR concordance[tiab] OR (intraclass[tiab] AND correlation*[tiab]) OR discriminative[tiab] OR "known group"[tiab] OR "factor analysis"[tiab] OR "factor analyses"[tiab] OR "factor structure"[tiab] OR "factor structures"[tiab] OR dimension*[tiab] OR subscale*[tiab] OR (multitrait[tiab] AND scaling[tiab] AND (analysis[tiab] OR analyses[tiab])) OR "item discriminant"[tiab] OR "interscale correlation*"[tiab] OR error[tiab] OR errors[tiab] OR "individual variability"[tiab]	9660087

	OR “interval variability”[tiab] OR “rate variability”[tiab] OR (variability[tiab] AND (analysis[tiab] OR values[tiab])) OR (uncertainty[tiab] AND (measurement[tiab] OR measuring[tiab])) OR “standard error of measurement”[tiab] OR sensitiv*[tiab] OR responsive*[tiab] OR (limit[tiab] AND detection[tiab]) OR “minimal detectable concentration”[tiab] OR interpretab*[tiab] OR ((minimal[tiab] OR minimally[tiab] OR clinical[tiab] OR clinically[tiab]) AND (important[tiab] OR significant[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR (small*[tiab] AND (real[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR “meaningful change”[tiab] OR “ceiling effect”[tiab] OR “floor effect”[tiab] OR “Item response model”[tiab] OR IRT[tiab] OR Rasch[tiab] OR “Differential item functioning”[tiab] OR DIF[tiab] OR “computer adaptive testing”[tiab] OR “item bank”[tiab] OR “cross-cultural equivalence”[tiab])	
#6	(“addresses”[Publication Type] OR “biography”[Publication Type] OR “case reports”[Publication Type] OR “comment”[Publication Type] OR “directory”[Publication Type] OR “editorial”[Publication Type] OR “festschrift”[Publication Type] OR “interview”[Publication Type] OR “lectures”[Publication Type] OR “legal cases”[Publication Type] OR “legislation”[Publication Type] OR “letter”[Publication Type] OR “news”[Publication Type] OR “newspaper article”[Publication Type] OR “patient education handout”[Publication Type] OR “popular works”[Publication Type] OR “congresses”[Publication Type] OR “consensus development conference”[Publication Type] OR “consensus development conference, nih”[Publication Type] OR “practice guideline”[Publication Type]) NOT (“animals”[MeSH Terms] NOT “humans”[MeSH Terms])	4244042
#7	#1 AND #2 AND #3 AND #4 AND #5	31186
#8	#7 NOT #6	2981

Search strategy for MEDLINE (Ovid)

Search time: 2021-5-3 14:46 (UTC+8)

Search	Query	Items found
#1	cancer.ab. or cancer.ti. or neoplasms.hw.	3022755
#2	patient?.ab. or patient?.ti. or survivor?.ab. or survivor?.ti. or patients.hw. or cancer survivors.hw. or survivors.hw.	6174913
#3	Cost.ab. or Cost.ti. or bill?.ab. or bill?.ti. or expense.ab. or expense.ti. or productivity loss.ab. or productivity loss.ti. or out-of-pocket.ab. or out-of-pocket.ti. or economic burden.ab. or economic burden.ti. or financial toxicity.ab. or financial toxicity.ti. or financial hardship.ab. or financial hardship.ti. or financial burden.ab. or financial burden.ti. or financial effect.ab. or financial effect.ti. or financial stress.ab. or financial stress.ti. or economic burden.ab. or economic burden.ti. or economic hardship.ab. or economic hardship.ti. or co-payment.ab. or co-payment.ti. or (cost of illness).hw.	419745
#4	Scale?.ab. or Scale?.ti. or patient reported outcome measur*.ab. or patient reported outcome measur*.ti. or PROM?.ab. or PROM?.ti. or measure*.ab. or measure*.ti. or Patient Reported Outcome Measures*.ab. or Patient Reported Outcome Measures*.ti. or (Surveys and Questionnaires).hw.	3676481
#5	(instrumentation or methods).fs.	4290317
#6	(Validation Studies or Comparative Study).pt.	1888426

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7	exp Psychometrics/	78410
8	psychometr*.ti,ab.	41677
9	(clinimetr* or clinometr*).tw.	1010
10	exp "Outcome Assessment Health Care"/	1188149
11	outcome assessment.ti,ab.	3533
12	outcome measure*.tw.	212018
13	exp Observer Variation/	43395
14	observer variation.ti,ab.	1008
15	exp Health Status Indicators/	319012
16	exp Reproducibility of Results/	416843
17	reproducib*.ti,ab.	143145
18	exp Discriminant Analysis/	11117
19	(reliab* or unreliab* or valid* or coefficient or homogeneity or homogeneous or internal consistency).ti,ab.	1217341
20	(cronbach* and (alpha or alphas)).ti,ab.	20027
21	(item and (correlation* or selection* or reduction*)).ti,ab.	19993
22	(agreement or precision or imprecision or "precise values" or test-retest).ti,ab.	339147
23	(test and retest).ti,ab.	24523
24	(reliab* and (test or retest)).ti,ab.	80342
25	(stability or interrater or inter-rater or intrarater or intra-rater or intertester or inter-tester or intratester or intra-tester or interobserver or inter-observer or intraobserver or intraobserver or intertechnician or inter-technician or intratechnician or intra-technician or interexaminer or inter-examiner or intraexaminer or intra-examiner or interassay or interassay or intraassay or intra-assay or interindividual or inter-individual or intraindividual or intra-individual or interparticipant or inter-participant or intraparticipant or intra-participant or kappa or kappa's or kappas or repeatab*).ti,ab.	507384
26	((replicab* or repeated) and (measure or measures or findings or result or results or test or tests)).ti,ab.	179804
27	(generaliza* or generalisa* or concordance).ti,ab.	78096
28	(intraclass and correlation*).ti,ab.	22856
29	(discriminative or known group or factor analysis or factor analyses or dimension* or subscale*).ti,ab.	506663
30	(multitrait and scaling and (analysis or analyses)).ti,ab.	132
31	(item discriminant or interscale correlation* or error or errors or individual variability).ti,ab.	252802
32	(variability and (analysis or values)).ti,ab.	90102
33	(uncertainty and (measurement or measuring)).ti,ab.	5260
34	(standard error of measurement or sensitiv* or responsive*).ti,ab.	1429839
35	((minimal or minimally or clinical or clinically) and (important or significant or detectable) and (change or difference)).ti,ab.	214078
36	(small* and (real or detectable) and (change or difference)).ti,ab.	6561
37	(meaningful change or ceiling effect or floor effect or Item response model or IRT or Rasch or Differential item functioning or DIF or computer adaptive testing or item bank or cross-cultural equivalence).ti,ab.	11890

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38	5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37	9103899
39	(child* or pediatric* or infan* or neonat* or newborn* or teen* or youth*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept, rare disease supplementary concept, unique identifier]	3139104
40	38 and 39	1017699
41	1 and 2 and 3 and 4 and 40	26

Search strategy for EMBASE (Ovid)

Search time: 2021-5-3 16:48 (UTC+8)

Search	Query	Items found
#1	cancer.ab. or cancer.ti. or neoplasms.hw.	2586704
#2	patient?.ab. or patient?.ti. or survivor?.ab. or survivor?.ti. or patients.hw. or cancer survivors.hw. or survivors.hw.	10409882
#3	Cost.ab. or Cost.ti. or bill?.ab. or bill?.ti. or expense.ab. or expense.ti. or productivity loss.ab. or productivity loss.ti. or out-of-pocket.ab. or out-of-pocket.ti. or economic burden.ab. or economic burden.ti. or financial toxicity.ab. or financial toxicity.ti. or financial hardship.ab. or financial hardship.ti. or financial burden.ab. or financial burden.ti. or financial effect.ab. or financial effect.ti. or financial stress.ab. or financial stress.ti. or economic burden.ab. or economic burden.ti. or economic hardship.ab. or economic hardship.ti. or co-payment.ab. or co-payment.ti. or (cost of illness).hw.	702534
#4	Scale?.ab. or Scale?.ti. or patient reported outcome measur*.ab. or patient reported outcome measur*.ti. or PROM?.ab. or PROM?.ti. or measure*.ab. or measure*.ti. or Patient Reported Outcome Measures*.ab. or Patient Reported Outcome Measures*.ti. or (Surveys and Questionnaires).hw.	5336565
#5	exp Intermethod comparison/ OR exp data collection method/ OR exp validation study/ OR exp feasibility study/ OR exp pilot study/ OR exp psychometry/ OR exp reproducibility/ OR reproducib*:ab,ti OR audit:ab,ti OR psychometr*:ab,ti OR clinimetr*:ab,ti OR clinometr*:ab,ti OR exp observer variation/ OR observer variation:ab,ti OR exp discriminant analysis/ OR exp validity/ OR reliab*:ab,ti OR valid*:ab,ti OR coefficient:ab,ti OR internal consistency:ab,ti OR (cronbach*:ab,ti AND (alpha:ab,ti OR alphas:ab,ti)) OR item correlation:ab,ti OR item correlations:ab,ti OR item selection:ab,ti OR item selections:ab,ti OR item reduction:ab,ti OR item reductions:ab,ti OR agreement:ab,ti OR precision:ab,ti OR imprecision:ab,ti OR precise values:ab,ti OR test-retest:ab,ti OR (test:ab,ti AND retest:ab,ti) OR (reliab*:ab,ti AND (test:ab,ti OR retest:ab,ti)) OR stability:ab,ti OR interrater:ab,ti OR inter-rater:ab,ti OR intrarater:ab,ti OR intra-rater:ab,ti OR intertester:ab,ti OR inter-tester:ab,ti OR intratester:ab,ti OR intratester:ab,ti OR interobeserver:ab,ti OR inter-observer:ab,ti OR intraobserver:ab,ti OR intraobserver:ab,ti OR intertechnician:ab,ti OR inter-technician:ab,ti OR intratechnician:ab,ti OR intratechnician:ab,ti OR interexaminer:ab,ti OR inter-examiner:ab,ti OR intraexaminer:ab,ti OR intraexaminer:ab,ti OR interassay:ab,ti OR inter-assay:ab,ti OR intraassay:ab,ti OR intra-assay:ab,ti OR interindividual:ab,ti OR inter-individual:ab,ti OR intraindividual:ab,ti OR intra-individual:ab,ti OR interparticipant:ab,ti OR inter-participant:ab,ti OR intraparticipant:ab,ti OR intraparticipant:ab,ti OR kappa:ab,ti OR kappas:ab,ti OR coefficient of variation:ab,ti OR	2110942

	repeatab*:ab,ti OR (replicab*:ab,ti OR repeated:ab,ti AND (measure:ab,ti OR measures:ab,ti OR findings:ab,ti OR result:ab,ti OR results:ab,ti OR test:ab,ti OR tests:ab,ti)) OR generaliza*:ab,ti OR generalisa*:ab,ti OR concordance:ab,ti OR (intraclass:ab,ti AND correlation*:ab,ti) OR discriminative:ab,ti OR known group:ab,ti OR factor analysis:ab,ti OR factor analyses:ab,ti OR factor structure:ab,ti OR factor structures:ab,ti OR dimensionality:ab,ti OR subscale*:ab,ti OR multitrait scaling analysis:ab,ti OR multitrait scaling analyses:ab,ti OR item discriminant:ab,ti OR interscale correlation:ab,ti OR interscale correlations:ab,ti OR (error:ab,ti OR errors:ab,ti AND (measure*:ab,ti OR correlat*:ab,ti OR evaluat*:ab,ti OR accuracy:ab,ti OR accurate:ab,ti OR precision:ab,ti OR mean:ab,ti)) OR individual variability:ab,ti OR interval variability:ab,ti OR rate variability:ab,ti OR variability analysis:ab,ti OR (uncertainty:ab,ti AND (measurement:ab,ti OR measuring:ab,ti)) OR standard error of measurement:ab,ti OR sensitiv*:ab,ti OR responsive*:ab,ti OR (limit:ab,ti AND detection:ab,ti) OR minimal detectable concentration:ab,ti OR interpretab*:ab,ti OR (small*:ab,ti AND (real:ab,ti OR detectable:ab,ti) AND (change:ab,ti OR difference:ab,ti)) OR meaningful change:ab,ti OR minimal important change:ab,ti OR minimal important difference:ab,ti OR minimally important change:ab,ti OR minimally important difference:ab,ti OR minimal detectable change:ab,ti OR minimal detectable difference:ab,ti OR minimally detectable change:ab,ti OR minimally detectable difference:ab,ti OR minimal real change:ab,ti OR minimal real difference:ab,ti OR minimally real change:ab,ti OR minimally real difference:ab,ti OR ceiling effect:ab,ti OR floor effect:ab,ti OR item response model:ab,ti OR irt:ab,ti OR rasch:ab,ti OR differential item functioning:ab,ti OR dif:ab,ti OR computer adaptive testing:ab,ti OR item bank:ab,ti OR cross-cultural equivalence:ab,ti	
#6	1 and 2 and 3 and 4 and 5	1721

Search strategy for CINAHL (EBSCO)

Search time: 2021-5-3 17:10 (UTC+8)

Search	Query	Items found
#1	TI cancer OR AB cancer OR MH neoplasms	438994
#2	AB patient? OR TI patient? OR AB survivor? OR TI survivor? OR MH patients OR MH “cancer survivors” OR MH survivors	1637009
#3	AB Cost OR TI Cost OR AB bill? OR TI bill? OR AB expense OR TI expense OR AB “productivity loss” OR TI “productivity loss” OR AB out-of-pocket OR TI out-of-pocket OR AB “economic burden” OR TI “economic burden” OR AB “financial toxicity” OR TI “financial toxicity” OR AB “financial hardship” OR TI “financial hardship” OR AB “financial burden” OR TI “financial burden” OR AB “financial effect” OR TI “financial effect” OR AB “financial stress” OR TI “financial stress” OR AB “economic burden” OR TI “economic burden” OR AB “economic hardship” OR TI “economic hardship” OR AB co-payment OR TI co-payment OR MH “cost of illness”	186320
#4	AB Scale? OR TI Scale? OR AB “patient reported outcome measur*” OR TI “patient reported outcome measur*” OR AB PROM? OR TI PROM? OR AB measure* OR TI measure* OR AB “Patient Reported Outcome Measures*” OR MH “Patient Reported Outcome Measures*” OR MH “Surveys and Questionnaires”	862405
#5	(MH “Psychometrics”) or (TI psychometr* or AB psychometr*) or (TI clinimetr* or AB	649386

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	clinimetr*) or (TI clinometr* OR AB clinometr*) or (MH "Outcome Assessment") or (TI outcome assessment or AB outcome assessment) or (TI outcome measure* or AB outcome measure*) or (MH "Health Status Indicators") or (MH "Reproducibility of Results") or (MH"Discriminant Analysis") or ((TI reproducib* or AB reproducib*) or (TI reliab* or AB reliab*) or (TI unreliab* or AB unreliab*)) or ((TI valid* or AB valid*) or (TI coefficient or AB coefficient) or (TI homogeneity or AB homogeneity)) or (TI homogeneous or AB homogeneous) or (TI "coefficient of variation" or AB "coefficient of variation") or (TI "internal consistency" or AB "internal consistency") or (MH "Internal Consistency+") or (MH "Reliability+") or (MH "Measurement Error+") or (MH "Content Validity+") or "hypothesis testing" or "structural validity" or "cross-cultural validity" or (MH "Criterion-Related Validity+") or "responsiveness" or "interpretability" or (TI reliab* or AB reliab*) and ((TI test or AB test) OR (TI retest or AB retest)) or (TI stability or AB stability) or (TI interrater or AB interrater) or (TI inter-rater or AB inter-rater) or (TI intrarater or AB intrarater) or (TI intra-rater or AB intrarater) or (TI intertester or AB intertester) or (TI inter-tester or AB inter-tester) or (TI intratester or AB intratester) or (TI intra-tester or AB intra-tester) or (TI interobserver or AB interobserver) or (TI inter-observer or AB inter-observer) or (TI intraobserver or AB intraobserver) or (TI intra-observer or AB intra-observer) or (TI intertechnician or AB intertechnician) or (TI inter-technician or AB inter-technician) or (TI intratechnician or AB intratechnician) or (TI intra-technician or AB intra-technician) or (TI interexaminer or AB interexaminer) or (TI inter-examiner or AB inter-examiner) or (TI intraexaminer or AB intraexaminer) OR (TI intra-examiner or AB intra-examiner) or (TI intra-examiner or AB intraexaminer) or (TI interassay or AB interassay) or (TI inter-assay or AB inter-assay) or (TI intraassay or AB intraassay) or (TI intra-assay or AB intra-assay) or (TI interindividual or AB interindividual) or (TI inter-individual or AB inter-individual) OR (TI intraindividual or AB intraindividual) or (TI intra-individual or AB intra-individual) or (TI interparticipant or AB interparticipant) or (TI inter-participant or AB inter-participant) or (TI intraparticipant or AB intraparticipant) or (TI intra-participant or AB intra-participant) or (TI kappa or AB kappa) or (TI kappa's or AB kappa's) or (TI kappas or AB kappas) or (TI repeatab* or AB repeatab*) or (TI responsive* or AB responsive*) or (TI interpretab* or AB interpretab*)	
#6	1 and 2 and 3 and 4 and 5	774

Search strategy for PsycINFO (EBSCO)

Search time: 2021-5-3 17:32 (UTC+8)

Search	Query	Items found
#1	TI cancer OR AB cancer OR MH neoplasms	1172
#2	AB patient? OR TI patient? OR AB survivor? OR TI survivor? OR MH patients OR MH "cancer survivors" OR MH survivors	16703
#3	AB Cost OR TI Cost OR AB bill? OR TI bill? OR AB expense OR TI expense OR AB "productivity loss" OR TI "productivity loss" OR AB out-of-pocket OR TI out-of-pocket OR AB "economic burden" OR TI "economic burden" OR AB "financial toxicity" OR TI "financial toxicity" OR AB "financial hardship" OR TI "financial hardship" OR AB "financial burden" OR TI "financial burden" OR AB "financial effect" OR TI "financial effect" OR AB "financial	3812

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	stress" OR TI "financial stress" OR AB "economic burden" OR TI "economic burden" OR AB "economic hardship" OR TI "economic hardship" OR AB co-payment OR TI co-payment OR MH "cost of illness"	
#4	AB Scale? OR TI Scale? OR AB "patient reported outcome measur*" OR TI "patient reported outcome measur*" OR AB PROM? OR TI PROM? OR AB measure* OR TI measure* OR AB "Patient Reported Outcome Measures*" OR MH "Patient Reported Outcome Measures*" OR MH "Surveys and Questionnaires"	47588
#6	1 and 2 and 3 and 4	33

Search strategy for Web of Science

Search time: 2021-5-3 17:57 (UTC+8)

Search	Query	Items found
#1	TI=cancer OR TS= (cancer OR neoplasms)	2596936
#2	TI=(patient? OR survivor?) OR TS=(patients OR "cancer survivors" OR survivors)	6121549
#3	TI=(Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress" OR "economic burden" OR "economic hardship") OR TS="cost of illness"	211496
#4	TI=(Scale? OR "patient reported outcome measur*" OR PROM? OR measure*) OR TS=("Patient Reported Outcome Measures*" OR "Surveys and Questionnaires")	788584
#6	#4 AND #3 AND #2 AND #1	425

Search strategy for ProQuest Dissertations and Theses

Search time: 2021-5-3 18:41 (UTC+8)

Search	Query	Items found
#1	ti(cancer OR neoplasms) OR su(cancer OR neoplasms)	41848
#2	Ti(patient? OR survivor? OR "cancer survivors") OR su(patient? OR survivor? OR cancer survivors)	44297
#3	Title : Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress" OR Summary : Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress" OR Title : "economic burden" OR "economic hardship" OR "cost of illness" OR Summary : "economic burden" OR "economic hardship" OR "cost of illness"	32997
#4	All : Scale? OR "patient reported outcome measur*" OR PROM? OR measure*	2738959
#6	#4 AND #3 AND #2 AND #1	3

Search strategy for Cochrane Library (Wiley)

Search time: 2020-5-18 13:57 (UTC+8)

Search	Query	Items found

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#1	(Cancer):ti,ab,kw	164548
#2	MeSH descriptor: [Neoplasms] explode all trees	81536
#3	(patient? OR survivor? OR "cancer survivors"):ti,ab,kw	1016991
#4	MeSH descriptor: [Patients] explode all trees	2715
#5	MeSH descriptor: [Cancer Survivors] explode all trees	255
#6	MeSH descriptor: [Survivors] explode all trees	1409
#7	(Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress" OR "economic burden" OR "economic hardship" OR "cost of illness"):ti,ab,kw	62550
#8	MeSH descriptor: [Cost of Illness] explode all trees	802
#9	(Scale? OR “patient reported outcome measur*” OR PROM? OR measure*):ti,ab,kw	560355
#10	MeSH descriptor: [Patient Reported Outcome Measures] explode all trees	706
#11	MeSH descriptor: [Surveys and Questionnaires] explode all trees	55522
#12	(“Validation Studies” OR “Comparative Study”):pt	167084
#13	MeSH descriptor: [Psychometrics] explode all trees	2846
#14	(psychometr*):ti,ab,kw	6346
#15	#1 OR #2	194269
#16	#3 OR #4 OR #5 OR #6	924360
#17	#7 OR #8 OR #9 OR #10 OR #11	621163
#18	#12 OR #13 OR #14	172446
#19	#15 AND #16 AND #17 AND #18	4304

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P3-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P5-6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P6-7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P6-7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P7-8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P7-8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P8
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P8-9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	P9
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Appendix 3
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Appendix 2 & Table 1
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Appendix 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Table 2
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P12
	23b	Discuss any limitations of the evidence included in the review.	P15-16
	23c	Discuss any limitations of the review processes used.	P15-16
	23d	Discuss implications of the results for practice, policy, and future research.	P16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	No amendments
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Title page
Competing interests	26	Declare any competing interests of review authors.	Title page
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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BMJ Open

Psychometric properties of self-reported financial toxicity measures in cancer survivors: A systematic review

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Primary Subject Heading:	Oncology
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Keywords:	ONCOLOGY, HEALTH ECONOMICS, Health economics < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Quality in health care < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Psychometric properties of self-reported financial toxicity measures in cancer survivors: A systematic review

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Psychometric properties of self-reported financial toxicity measures in cancer survivors: A systematic review

Abstract

Objective: The aim of this systematic review was to summarize the psychometric properties of patient-reported outcome measures (PROMs) measuring financial toxicity (FT) in cancer survivors.

Design: This systematic review was conducted according to the guidance of the Consensus-Based Standards for the Selection of Health Measurement Instruments (COSMIN) methodology.

Data sources: Comprehensive searches were performed in PubMed, MEDLINE, Embase, CINAHL, PsycINFO, Web of Science, ProQuest, and Cochrane Library from database inception to February 2022.

Eligibility criteria for selecting studies: We included studies that reported any PROMs for measuring FT in cancer survivors who were ≥18 years old. FT was defined as perceived subjective financial distress resulting from objective financial burden. Studies that were not validation studies and that used a PROM only as an outcome measurement were excluded.

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Data extraction and synthesis: Two reviewers independently extracted data from the included papers. We used the COSMIN criteria to summarize and evaluate the psychometric properties of each study regarding structural validity, internal consistency, reliability, measurement error, hypothesis testing for construct validity, cross-cultural validity/measurement invariance, criterion validity, and responsiveness.

Results: A total of 23 articles (21 PROMs) were eligible for inclusion in this study. The findings highlighted that the Comprehensive Score for Financial Toxicity (COST) had an adequate development process and showed better psychometric properties than other PROMs, especially in internal consistency (Cronbach's α=0.92), reliability (intraclass correlation coefficient=0.80), and hypothesis testing (r=0.42-0.20).

Conclusions: From a psychometric property perspective, the COST could be recommended as the most suitable worldwide available measure for use in research and clinical practice across different contexts. We suggest that PROMs should be selected only after careful consideration of the local socioeconomic context. Future studies are warranted to develop various FT PROMs based on different social and cultural backgrounds and to clarify the theoretical grounds for assessing FT.

Keywords: cancer, financial toxicity, measurement, PROM, systematic review

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Strength and limitation of this study

- This is the first systematic review that comprehensively summarized the psychometric properties of 21 PROMs evaluating financial toxicity in cancer survivors.
- The results may provide quantitative evidence for researchers and healthcare professionals to choose PROMs measuring cancer survivors' financial toxicity in future scientific research and clinical practice.
- This review only included studies that aimed to evaluate the measurement properties of financial toxicity PROMs.

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Introduction

The rising cost associated with advancements in cancer treatment and lengthening of cancer survivorship poses a significant challenge to survivors, caregivers, and public healthcare systems.[1,2] Total global spending on cancer medications grows at a compound annual growth rate of 6.5%, growing from US$ 96 billion in 2013 to US$ 173 billion in 2020, which is nearly twice the rate of global gross domestic product (GDP) growth.[3-5] The majority of cancer survivors in middle-income and low-income countries/regions depend on out-of-pocket payments, which may lead to global inequalities in healthcare expenditures and financial insecurity for vulnerable groups.[6,7]

The term "financial toxicity (FT)" has been described as the economic effect of cancer treatment in the age of precision medicine.[2,8,9] Witte and colleagues described FT as "the patient-reported outcome (PRO) of perceived subjective financial distress resulting from objective financial burden".[10] This concept covers both the objective financial burden and the subjective financial distress that cancer survivors face as a result of high out-of-pocket medical expenses. Regarding the terminology, "financial toxicity", "financial burden" and "financial distress" are often used interchangeably in research and share a similar definition.[10,11] In this review, the authors agreed to consistently use the term "financial toxicity." Financial toxicity is usually measured by patient-reported outcome measures (PROMs); choosing a PROM with high validity and reliability is a prerequisite for robust results.

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There are a few cancer-specific and generic FT PROMs that have been reported and used in different contexts. As one of the recent cancer-specific FT PROMs, the Comprehensive Score for Financial Toxicity (COST) is the most commonly used measure for assessing FT.[12] In addition to COST, other cancer-specific measures have been widely used, including the Breast Cancer Finances Survey Inventory (BCFS),[13] Socioeconomic Wellbeing Scale (SWBS),[14] and InCharge Financial Distress/Financial Wellbeing Scale (InCharge).[15] Additionally, validated subscales, such as the Social Difficulties Inventory Cancer Care Outcomes (SDI), the Cancer Care Outcomes Research and Surveillance Consortium patient survey (CanCORS), and Italian version of the Edmonton Symptom Assessment System (ESAS)-Total Care (TC), were also used to evaluate FT.[16-18] However, existing PROMs vary significantly in their state of development and degree of validation, and many PROMs have not been psychometrically tested.

A preliminary literature search was conducted in PubMed, PsycINFO (EBSCO), Cochrane Library (Wiley) and JBI (Ovid), which revealed that there exist some reviews regarding measures of FT. Witte and colleagues summarized the content of 352 items from 34 studies measuring FT in cancer survivors.[10] However, this review did not report the psychometric properties of the included PROMs, and most of the included PROMs were not validated through a scientific process, which made it difficult for readers to choose the best measure from existing PROMs to evaluate the

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level of FT. Salman and colleagues conducted a systematic review and found 8 PROMs and 2 caregiver-reported measures for assessing financial burden in adolescents and young adults.[19] However, this review focused only on PROMs assessing FT in adolescents and young adults with cancer. The psychological properties of FT measures in adult cancer survivors are still unknown.

The reproducibility, reliability, and accuracy of PROMs are the fundamental premise for achieving robust results. Therefore, it is necessary to summarize the psychometric properties of existing PROMs for future research. However, this information is still lacking. The aim of this systematic review was to summarize the psychometric properties of PROMs for measuring FT in cancer survivors. The review was conducted according to the guidance of the Consensus-Based Standards for the Selection of Health Measurement Instruments (COSMIN) methodology and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[20,21] The protocol of this review was published in *BMJ OPEN* in 2020.[22] The registration number of the protocol in PROSPERO was CRD42021254721.

Methods

Search strategy

First, we conducted a limited search via PubMed to capture keywords from which to develop search strategies for each database. Subsequently, all identified search strategies across databases were performed in PubMed/MEDLINE, MEDLINE

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(Ovid), Embase (Ovid), CINAHL (EBSCO), PsycINFO (EBSCO), Web of Science, ProQuest Dissertations and Theses, and Cochrane Library (Wiley). The search time frame was set from database inception to February 2022. To include more studies published in 2021 and 2022, the end date of the search was updated to February 2022.[22] In PubMed/Medline, we searched papers in English using MeSH terms ([cancer OR neoplasms] AND [“cancer survivors” OR patient OR survivors] AND “cost of illness”) combined with (cancer OR [patient* OR survivor*] AND [cost OR bill* OR expense OR productivity loss OR “out-of-pocket” OR “economic burden” OR “financial toxicity” OR “financial hardship” OR “financial burden”]). The COSMIN measurement properties filter and exclusion filter were also utilized in the search box. The search strategies for each database are presented in Appendix 1. Finally, the references of all included studies were manually reviewed to supplement the database search.

Inclusion and exclusion criteria

The inclusion criteria were as follows: 1) studies that reported any PROMs for measuring FT in cancer survivors who were ≥18 years old. If the studies reported results in a population combined with both ≥18- and <18-year-old cancer survivors and the majority of survivors were not < 18 years old, the studies were also considered; 2) studies that evaluated at least one measurement property; and 3) studies published in English. The exclusion criteria were as follows: 1) studies that were not validation studies and used a PROM only as an outcome measurement; 2) studies that

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used a PROM as a comparator for another instrument; 3) studies that did not provide empirical data; and 4) if a measure was a quality of life PROM and had a domain that assessed FT, we included only the original version of the PROM. If the measure/domain included only one item and reported the measurement property as an independent domain, the measure/domain was also considered.

Study screening and selection

We imported all identified citations by search strategies into Endnote X8 (Clarivate Analytics, PA, USA). After duplicates were removed, two reviewers (ZZ & WX) independently screened all titles, abstracts, and full texts (ZZ & WX) based on the established inclusion and exclusion criteria. Any disagreements were resolved by a third reviewer (YH).

Quality appraisal

Two reviewers (WH & YS) assessed the methodological quality of the PROM of the included studies by using the COSMIN Risk of Bias Checklist (Appendix II).[19] The checklist consisted of 10 domains (116 items), including PROM development, content validity, structural validity, internal consistency, cross-cultural validity, reliability, measurement error, criterion validity, hypothesis testing, and responsiveness. Each measurement property was rated as “very good”, “adequate”, “doubtful” or “inadequate quality”. According to the COSMIN guidelines, the methodological quality of a single study is rated based on the worst score count

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method. For example, if the lowest rating is "inadequate" in the PROM development domain, the overall methodological quality of that domain is "inadequate". The worst score counts method takes into account that inadequate quality items could affect the overall results of the measurement property of each PROM. Any discrepancies were resolved by a third reviewer (ZZ).

Data extraction

Two reviewers (ZZ & WX) independently extracted data from the included papers, including authors, year of publication, PROM, country/language, study design, target population, sample size, domains, number of items, total score range, and main findings. The main findings regarding psychometric properties, including content validity, structural validity, internal consistency, cross-cultural validity, reliability, measurement error, criterion validity, hypothesis testing, and responsiveness, were also extracted. Any discrepancies were resolved through discussion between the two reviewers.

Data synthesis

We used the COSMIN criteria to summarize and evaluate the psychometric properties of each study regarding structural validity, internal consistency, reliability, measurement error, hypothesis testing for construct validity, cross-cultural validity/measurement invariance, criterion validity, and responsiveness. Each measurement property from each study was rated as sufficient (+), insufficient (−), or

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indeterminate (?). The criteria for the measurement property rating can be found in Appendix II. If the ratings of one psychometric property per study were all sufficient (+) or insufficient (-), the results were pooled, and the overall rating was rated as sufficient (+) or insufficient (-). If the ratings were inconsistent, explanations of inconsistency were explored (e.g., different languages). For example, in our review, different language, social, economic, and cultural contexts may contribute to inconsistencies in psychometric properties. Our review team (ZZ, WJ, HW, and YS) discussed the potential explanations of inconsistency. If the review team regarded the explanation as reasonable, we provided ratings ("+", "-", and "?") in subgroups (e.g., language subgroup). If the explanation was not reasonable, the overall rating of this measurement property was rated as inconsistent (**±**).

Assessing certainty of evidence

We used a modified Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system to assess the certainty of evidence.[19] Each piece of evidence was graded for risk of bias, inconsistency, imprecision, and indirectness. The instructions for downgrading for risk of bias, inconsistency, imprecision, and indirectness are shown in Appendix II. Four reviewers (ZZ, WJ, HW, and YS) independently assessed the grade. Any discrepancies were resolved by discussion.

Patient and public involvement

No patients or the public were directly involved in the development of the

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research question, selection of the outcome measures, design and implementation of the study, or interpretation of the results.

Results

Literature search

Figure 1 shows the process of literature screening and selection. A total of 9399 articles were identified via databases. Six articles were found by additional supplementary searches. After duplications were removed, a total of 11731 articles were retained, 11669 articles were deleted after reading the title and abstract, and 39 were deleted after full-text reading. Finally, a total of 23 articles (21 PROMs) were eligible for inclusion in this study.[12,14,16,23-42]

Study description

Table 1 shows the characteristics of the included studies. All included studies were published from 2005 to 2022. Eight studies were conducted in the United States,[12,14,23,27,30,37,39,41] four in the United Kingdom,[16,29,35,38] two in Canada,[31,36] and two in China (mainland and Hong Kong),[25,39] India,[26,34] and Italy.[33,42] One study was conducted in 12 countries in Europe and North America.[22,23] Other studies were conducted in Brazil[32] and Iran[34]. A total of 12362 participants were included, ranging from 7[36] to 5901[41] per study. The majority of studies assessed FT in multiple types of cancer. Only two studies focused on a single type of cancer, namely, lung, colorectal, or head and neck cancer. [31,37]

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Among the 21 PROMs, 7 were FT-related domains of quality of life PROMs, and 14 were independent PROMs focusing on FT. All PROMs were validated in cancer survivors. Fifteen PROMs were in English,[12,14,16,23,25-31,35,37,38,40,41,42] and two were in Chinese.[24,39] Other languages included French,[36] Portuguese,[32] Italian,[33,42], Hindi,[25,26] and Persian.[34] The number of items evaluating FT ranged from 3[40] to 23.[36] The French version of the Patient Self-Administered Financial Effects Questionnaire (P-SAFE) did not report the total score range of the whole PROM.[36]

Quality assessment

Methodological quality assessment. Table 2 shows the methodological quality of the 23 included studies by using the COSMIN checklist. In the PROM development domain, only one study was rated as adequate,[42] three studies were rated as doubt,[12,24,27,29] and the others were rated as inadequate. Two studies reported adequate information in testing the relevance, comprehensiveness, and comprehensibility of PROMs.[12,27,29] One study reported adequate relevance and comprehensiveness.[42] Among all studies, the most reported domain was internal consistency, except one study.[36] Limited information could be retrieved on cross-cultural validity (three studies),[31,32,36] criterion validity (six studies),[16,23,33,35,38,40] reliability (ten studies),[12,16,24,27,28,33,35,38,39,40,42] and responsiveness (two studies).[31,39] No data were identified as measurement error.

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Measurement property assessment. Table 3 shows the quality of the psychometric properties retrieved from 21 PROMs. Only the Persian version of the COST-v2 and Subjective Financial Distress Questionnaire (SFDQ) were rated as "+" in structural validity.[26,34] There were 17 PROMs rated as "+" in internal consistency.[12,14,16,23,24,26-29,31,32,34,35,37-39,41,42] Eight PROMs were rated as "+" in reliability.[12,24,26-29,31,35] Ten PROMs were rated as "+" in hypothesis testing.[12,14,23,24,27-31,33,35,39] Limited information was retrieved on cross-cultural validity (two PROMs),[32,36] criterion validity (six PROMs),[16,24,33,35,38,40] and responsiveness (two PROMs).[31,39] No PROMs reported data on measurement error.

Certainty of evidence

Table 4 shows the certainty of evidence for each measurement property. Among all included PROMs, the COST showed the best psychometric properties compared to other measures. The COST and its seven versions were rated as having high evidence of structural validity, internal consistency, hypothesis testing, and criterion validity.[12,24,25,27,28,32,33,34,39] The Financial Index of Toxicity (FIT) and Impact of Cancer – Childhood Survivors (IOC-CS) financial problems domain reported data on five properties and were rated on a scale from "very low evidence" to "high evidence."[31,40]

Discussion

This systematic review identified 21 PROMs and domains of PROMs evaluating

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FT in cancer survivors, including the COST (original, Brazilian, India, Italian, Persian, Simplified Chinese, Traditional Chinese version), FIT, Personal Financial Burden (PFB), P-SAFE, SWBS, Quality of Life in Adult Cancer Survivors (QLACS) financial problems domain, Chronic Cancer Experiences Questionnaire (CCEQ) financial advice domain, Patient-Reported Outcome for Fighting Financial Toxicity (PROFFIT), Patient Roles and Responsibilities Scale (PRRS) financial well-being domain, SDI-21 providing for the family domain, SDI-16 money matters domain, Subjective Financial Distress Questionnaire (SFDQ), IOC-CS financial problems domain, and Cancer Problems in Living Scale (CPILS) employment/financial domain. Overall, the COST had a complete development process compared to other PROMs and showed the best psychometric properties, especially in terms of internal consistency, reliability, and hypothesis testing. To the best of our knowledge, this is the first systematic review that has summarized the psychometric properties of FT PROMs in cancer survivors and reported the certainty of evidence for each property of PROMs. The results may provide quantitative evidence for researchers and healthcare professionals to choose PROMs measuring cancer survivors' FT in future scientific research and clinical practice.

The results highlighted that the COST (of which we studied both version 1 and version 2) had better psychometric properties than other specific and generic PROMs in terms of internal consistency, reliability, and hypothesis testing. The COST could be recommended as the most suitable worldwide available measure for use in research

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and clinical practice across different contexts. Other systematic reviews have also suggested that the COST is a promising measure from a content perspective[10,11]. From a psychometric standpoint, there are a few issues that one must face when evaluating financial toxicity in cancer survivors using the COST. First, caution should be taken when using the COST in different socioeconomic conditions outside the United States. In some countries in Europe or Asia, the majority of medical expenses are covered by social health insurance, and direct out-of-pocket payments are replaced by prepayment from health insurance contributions.[43,44] In addition, social security systems can benefit cancer survivors who are not able to work.[45] These two socioeconomic factors may affect cancer survivors' understanding regarding some items related to medical spending and indirect cost. However, few COST validation studies have considered socioeconomic issues, adapted the measure in a local context, or provided data on cross-cultural validity. It is recommended that future COST validation studies recruit cancer survivors across multiple social and cultural backgrounds to assess cross-cultural measurement invariance.

Second, the original construct and item generation for the COST were based on a literature search; thus, the theoretical grounds for the measure are unclear, and the instrument may not capture detailed information related to the construct. Theoretical frameworks and conceptual models are crucial for self-reported measures to capture subtle changes in constructs.[46] Although FT is a relatively new concept, certain models can guide item generation in the development of future FT PROMs.

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Tucker-Seeley and colleagues developed a conceptual model of FT and emphasized three components of financial burden, namely, the material, psychosocial, and behavioral domains.[47] Head developed SWBS based on James Coleman's Theory of Social Class; this scale contains 17 items across 3 domains: human capital, material capital, and social capital.[14,30,48] Witte and colleagues' systematic review analyzed 352 different questions regarding financial spending and found six domains (financial spending, financial resources, psychosocial affect, support seeking, coping care, and coping lifestyle) that can represent reactions to subjective financial distress.[10] Other theories and models, including the Wreckers theory of financial distress, ecological theory, and the functionalist tradition, have also been widely used in cancer survivors.[49-51] With the increasing number of theoretical studies related to FT, the theoretical grounds for future PROMs need to be clarified.

In addition to the COST, two other PROMs, namely, the FIT and the IOC-CS financial problems domains, also provided adequate data on psychometric properties. The FIT is relatively new and has fewer items than the other included measures. This measure was developed by Hueniken and colleagues and has been validated only in survivors with head and neck cancer.[31] Head and neck cancer, especially laryngeal and hypopharyngeal cancer, has particularly large impacts on survivors' daily function (e.g., speech and eating) after treatment and affects survivors' ability to return to work.[52,53] Only 32% to 59% of head and neck cancer survivors return to work after treatment.[54] This form of cancer also has short- and long-term financial

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consequences for caregivers and their families.[55] Therefore, future studies should be aware that the FIT may not be directly applicable to other cancer populations.

Regarding PROM development, we found that only two PROMs, PROFFIT and SFDQ, were not developed in the context of English-speaking developed countries such as the United States, the United Kingdom, and Canada. The socioeconomic contexts and healthcare systems in these countries may be significantly different from those in other parts of the world and ultimately lead to a nuance in the perceived causes and consequences of FT. Previous studies have reported that FT is closely related to broad social determinants of economic circumstances. Factors including healthcare policy, healthcare system, insurance system, specific micro contexts, and the level of regional economic development could not only affect the cancer survivors' perceived level of FT but also determine the origins of FT.[56,57] Additionally, cultural factors (e.g., a cultural emphasis on saving and a cultural imperative to have a large family) also affect cancer survivors' perceived financial security and economic burden.[58]

PROFFIT, which was developed in 2021 in the Italian context, also reported higher quality PROM development and content validity than other PROMs. We would consider it to be a good FT PROM against the COSMIN criteria if more validation studies were conducted to report a greater effect size of the measurement properties. Therefore, we recommend that researchers use context-specific measures

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to assess FT in cancer survivors (e.g., using PROFFIT in Italy). Further studies are warranted to develop various FT PROMs based on different social and cultural backgrounds. Worldwide measures, such as COST, should be analyzed to determine the differences between social, cultural, and economic contexts.

Limitations

We acknowledge that there are some limitations to this study. First, this review included only studies that aimed to evaluate the measurement properties of FT PROMs. Many studies that aimed to explore the level of FT in cancer survivors also reported the reliability and validity of PROMs. Therefore, the PROMs we summarized in this systematic review had higher psychometric quality than other measures that we did not list in this review. Second, we included only studies published in English. Therefore, studies published in other languages were not included, which may affect the conclusion of this review. Third, we included only the original version of the FT domain from PROMs assessing quality of life in cancer survivors, such as the EORTC QLQ-C30 and the QLACS. Over 20 language versions of these PROMs do not provide sufficient details on the FT domain individually.

Conclusion

This systematic review summarized the psychometric properties of 20 PROMs evaluating FT in cancer survivors. The findings highlighted that, from a psychometric property perspective, the COST had an adequate PROM development process and

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showed the best psychometric properties among all examined PROMs, especially in internal consistency, reliability, and hypothesis testing; thus, we recommend the COST as the most suitable worldwide available measures for use in research and clinical practice across different contexts. The FIT and the IOC-CS financial problems domain also had adequate psychometric properties. We suggest that PROMs should be selected only after careful consideration of the local socioeconomic context. Future studies are warranted to develop various FT PROMs based on different social and cultural backgrounds and a clear theoretical basis for assessing FT.

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Contributors

ZZ and WX designed the systematic review, conducted data searching, extraction and analysis, assessing the certainty of evidence, and wrote the draft of the manuscript. HW and YS conducted quality appraisal and assessing the certainty of evidence. WS, LL, JP, and YH provided critical comments. All authors approved the final version of the manuscript.

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Competing interests

None.

Patient and public involvement

Patients and the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Patient consent for publication

Not applicable.

Ethics approval

This systematic review did not require ethical approval.

Data available statement

No additional data are available.

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Table 1 Overview of the included studies

Author (year), country	PROM	Country	Language(s) of PROM	Study design	Target population	Sample size	Measurement domain	Number of items	Total score range
Avis et al., 2005	QLACS financial problems domain	US	English	Development and validation study	Age: 71.4±11.5 yrs Male: 42% Cancer type: Breast, bladder, head and neck, gynecologic, prostate, colorectal cancer	242	1 domain regarding financial problems	4	4-28
Chan et al., 2021	COST-v2, *Traditional Chinese version*	Hong Kong	Traditional Chinese	Validation study	Age: 59.9±11.1 yrs Male: 35.3% Cancer type: Breast, gynecological, head and neck, gastric and colorectal, genitourinary, lung, hematological, skin, bone and soft tissue, brain and central nervous system cancer and others	640	No subdomain	12	0-44
Dar et al., 2021	COST-v1, *India version*	India	Hindi or English	Validation study	Age: 49.5±16.8 yrs Male: 82.8% Cancer type: Tongue, gingival buccal sulcus, buccal mucosa, supraglottic larynx, hypopharynx, parotid and others	29	No subdomain	11	0-44
Dar et al., 2021	SFDQ	India	Indian or English	Development and validation study	Age: 18-59 68.3%; ≥60 31.7% Male: 85.9% Cancer type: head and neck cancer	142	Five domains: financial resources; financial spending; psychosocial affect; coping care & coping lifestyles; support	14	0-28

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							seeking		
de Alcantara Nogueira et al., 2020	COST-v1, *Brazilian version*	Brazil	Brazilian Portuguese	Validation study	Mean age: 56 yrs Male: 40.5% Cancer type: Not specific	126	No subdomain	11	0-44
De Souza et al., 2014 De Souza et al., 2017	COST-v1	US	English	Development study and validation study	Age: 58.4±11.5 yrs Male: 41.6% Cancer type: Not specified (diagnosis of AJCC stage IV cancer)	233	No subdomain	11	0-44
Durber et al., 2021	COST-v1	Australia	English	Validation study	Age: ≤50 yrs 23%; 51-64 yrs 30%;≥65 yrs 48% Male: 46% Cancer type: Thoracic, breast, sarcoma, skin, central nervous system, gynecological, head and neck, colorectal, upper gastrointestinal, urological, and miscellaneous cancer	257	No subdomain	11	0-44
Harley et al., 2019	CCEQ financial advice domain	UK	English	Development and validation study	*Pilot study:* Age: 65 (41-90) yrs Male: 48.5% Cancer type: Breast, colorectal/gastrointestinal, gynecological, prostate, and renal cancer *Final study:* Age: 67 (41-88) yrs Male: 50.0% Cancer type: Breast, colorectal/gastrointestinal, gynecological, prostate, and renal cancer	103 for pilot study 313 for final study	1 domain regarding financial advice	5	5-25

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Head, 2008 Head & Faul, 2008	SWBS	US	English	Development and validation study	Age: 59.6±12.7 yrs Male: 35.7% Cancer type: Breast, melanoma, head and neck, prostate, rectum/anus, colon, endometrium, lung/tracheal/bronchus, and non-Hodgkin's lymphoma	266	Two domains: material and social capital.	17	0-68
Hueniken et al., 2020	FIT	Canada	English	Development and validation study	Age: 61.6 (25.5-88.5) yrs Male: 77.2% Cancer type: Oropharynx, oral cavity, larynx, nasopharynx, hypopharynx cancers, and others	430	No subdomain	9	0-100
Ripamonti et al., 2020	COST-v2, *Italian version*	Italy	Italian	Validation study	Age: 61.5±12.7 yrs Male: 52.5% Cancer type: Breast, lung, colon, gastric, hepatocellular, endometrial, prostate, sarcoma, bladder, head and neck, Hodgkin lymphoma, non-Hodgkin lymphoma, leukemia, myeloma, and others	118	No subdomain	11	0-44
Riva et al., 2021	PROFFIT	Italy	Italian or English	Development and validation study	Age: 29-82 yrs Male: 41.3% Cancer type: breast, lower gastrointestinal tract, genitourinary, thoracic, upper gastrointestinal tract, and others	184	No subdomain	7 outcome items and 8 determinant items	0-100
Sharif et al., 2020	COST-v2, *Persian version*	Iran	Persian	Validation study	Age: 50.0±14.3 yrs Male: 51.0% Cancer type: Not specific	398	No subdomain	11	0-44
Shilling et al., 2018	PRRS financial wellbeing domain	UK	English	Development and validation study	Age: ≤50 yrs 25%; 51-65 yrs 41%; ≥66 yrs 34% Male: 23% Cancer type: breast, gynecological, lung, and melanoma cancers	135	1 domain regarding financial wellbeing	6	0-24

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Tremblay et al., 2020	P-SAFE, *French Version*	Canada	French	Cross-adaption study	Age: 50-59 yrs 57%; 60-69 yrs 29%; ≥70 yrs 34% Male: 14% Cancer type: colorectal, lung, breast, and prostate cancer.	7	NR	23	NR
Veenstra et al., 2014	PFB	US	English	Validation study	Age: <50 yrs 17%; 50-64 yrs 37%; 65-74 yrs 23%; >75 yrs 24% Male: 53% Cancer type: Stage III colorectal cancer	956	No subdomain	7	0-7
Wright et al., 2005	SDI-21 providing for the family domain	UK	English	Development and validation study	Age: 53.8±14.1 yrs Male: Not specific Cancer type: brain, lung cancers and others	271	1 domain regarding providing for the family	5	0-20
Wright et al., 2011	SDI-16 money matters domain	UK	English	Development and validation study	Age: 56 (18-88) yrs for men; 56 (21-88) yrs for women Male: 48% Cancer type: breast, gastrointestinal, hematology, gynecological, germ cell, head and neck, lung, genitourinary, and others	652	1 domain regarding money matters	5	0-20
Yu et al., 2021	COST-v1, *Simplified Chinese version*	Mainland China	Chinese	Validation study	Age: 57.0±9.2 yrs Male: 45.7% Cancer type: lung, stomach, colorectal, and breast cancer	440	No subdomain	11	0-44
Zebrack et al., 2010	IOC-CS financial problems domain	US	English	Validation study	Age: 26.7±5.3 yrs Male: 48.0% Cancer type: hematological, brain, and solid tumors/soft tissue tumors	519	1 domain regarding financial problems	3	1-15
Zhao et al., 2009	CPILS employment/financial domain	US	English	Validation study	Age: ≤55 yrs 48.8%; >55 51.2% Male: 41.6% Cancer type: breast, prostate, colorectal, bladder, uterine, kidney, lung, and ovarian cancer;	5901	1 domain regarding employment/finances	6	0-12

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			melanoma of skin; non-Hodgkin lymphoma,

CCEQ: Chronic Cancer Experiences Questionnaire; COST: Comprehensive Score for Financial Toxicity; CPILS: Cancer Problems in Living Scale; FIT: Financial Index of Toxicity; IOC-CS: Impact of Cancer-Childhood Survivors; PFB: Personal Financial Burden; PROFFIT: Patient-Reported Outcome for Fighting Financial Toxicity; PROM: Patient-Reported Outcome Measures; PRRS: Patient Roles and Responsibilities Scale; P-SAFE: Patient Self-Administered Financial Effects questionnaire; QLACS: Quality of Life in Adult Cancer Survivors; SDI: Social Difficulties Inventory Cancer Care Outcomes; SFDQ: Subjective Financial Distress Questionnaire; SWBS: Socioeconomic Wellbeing Scale

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629 **Table 2** Methodological quality assessment of the measures

Author (year)	PROM	Measurement property: methodological quality by study									
		PROM development	Content validity	Structural validity	Internal consistency	Cross-cultural validity	Reliability	Measurement error	Criterion validity	Hypothesis testing	Responsiveness
Avis et al., 2005	QLACS financial problems domain	Inadequate	R: NR C1: NR C2: NR	Adequate	Very good	NR	NR	NR	Adequate	Doubtful	NR
Chan et al., 2021	COST-v2, *Traditional Chinese version*	Doubtful	R: Doubtful C1: NR C2: Doubtful	Very good	Very good	NR	Doubtful	NR	NR	Very good	NR
Dar et al., 2021	COST-v1, *India version*	Inadequate	R: NR C1: NR C2: NR	Inadequate	Very good	NR	NR	NR	NR	NR	NR
Dar et al., 2021	SFDQ	Inadequate	R: Adequate C1: NR C2: NR	Very good	Very good	NR	NR	NR	NR	NR	NR
de Alcantara Nogueira et al., 2020	COST-v1, *Brazilian version*	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	Inadequate	NR	NR	NR	NR	NR
De Souza et al., 2014 De Souza et al., 2017	COST-v1	Doubtful	R: Adequate C1: Adequate C2: Adequate	NR	Very good	NR	Adequate	NR	NR	Very good	NR
Durber et al., 2021	COST-v1, *Australia version*	Inadequate	R: NR C1: NR C2: NR	NR	Very good	NR	Adequate	NR	NR	Very good	NR

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Harley et al., 2019	CCEQ financial advice domain	Doubtful	R: Adequate C1: Adequate C2: Adequate	Adequate	Very good	NR	NR	NR	NR	Very good	NR
Head, 2008 Head & Faul, 2008	SWBS	Inadequate	R: NR C1:Doubt C2: NR	Very good	Very good	NR	NR	NR	Very good	Very good	NR
Hueniken et al., 2020	FIT	Inadequate	R: NR C1: NR C2: NR	Adequate	Very good	Doubtful	NR	NR	NR	Very good	Very good
Ripamonti et al., 2020	COST-v2, *Italian version*	Inadequate	R: NR C1: NR C2: NR	Inadequate	Very good	NR	Inadequate	NR	Very good	Very good	NR
Riva et al., 2021	PROFFIT	Adequate	R: Adequate C1: Adequate C2: NR	Adequate	Very good	NR	Adequate	NR	NR	NR	NR
Sharif et al., 2020	COST-v2, *Persian version*	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	NR	NR	NR	NR	Inadequate	NR
Shilling et al., 2018	PRRS financial wellbeing domain	Inadequate	R: NR C1: NR C2: NR	Adequate	Very good	NR	Inadequate	NR	Inadequate	Inadequate	NR
Tremblay et al., 2020	P-SAFE, *French Version*	Inadequate	R: NR C1: NR C2: NR	NR	NR	Inadequate	NR	NR	NR	NR	NR
Veenstra et al., 2014	PFB	Inadequate	R: NR C1: NR C2: NR	Very good	Inadequate	NR	NR	NR	NR	NR	NR
Wright et al., 2005	SDI-21 providing for the family domain	Inadequate	R: Doubt C1: NR C2: NR	Very good	Very good	NR	NR	NR	NR	NR	NR
Wright et al., 2011	SDI-16 money matters	Inadequate	R: Doubt C1: NR	Very good	Very good	NR	Inadequate	NR	Inadequate	NR	NR

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	domain		C2: NR								
Yu et al., 2021	COST-v1, *Simplified Chinese version*	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	NR	Very good	NR	NR	Doubt	Doubt
Zebrack et al., 2010	IOC-CS financial problems domain	Inadequate	R: NR C1: NR C2: Doubt	Adequate	Very good	NR	Adequate	NR	Inadequate	Very good	NR
Zhao et al., 2009	CPILS employment/financial domain	Inadequate	R: NR C1: NR C2: NR	Very good	Very good	NR	NR	NR	NR	Very good	NR

CCEQ: Chronic Cancer Experiences Questionnaire; COST: Comprehensive Score for Financial Toxicity; CPILS: Cancer Problems in Living Scale; FIT: Financial Index of Toxicity; IOC-CS: Impact of Cancer-Childhood Survivors; NA: Not Applicable; NR: Not Report; PFB: Personal Financial Burden ; PROFFIT: Patient-Reported Outcome for Fighting Financial Toxicity; PROM: Patient-Reported Outcome Measures; PRRS: Patient Roles and Responsibilities Scale ; P-SAFE: Patient Self-Administered Financial Effects questionnaire; QLACS: Quality of Life in Adult Cancer Survivors; SDI: Social Difficulties Inventory Cancer Care Outcomes; SFDQ: Subjective Financial Distress Questionnaire; SWBS: Socioeconomic Wellbeing Scale

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Table 3 Rating of measurement properties

PROM	Author (year)	Structural validity	Internal consistency	Reliability	Measurement error	Hypothesis testing	Cross-cultural validity	Criterion validity	Responsiveness
CCEQ financial advice domain	Harley et al., 2019	- (no data)	+ (Cronbach's α= 0.71–0.88)	NR	NR	+ (r=0.35-0.39)	NR	NR	NR
COST-v1	De Souza et al., 2014; De Souza et al., 2017	NR	+ (Cronbach's α=0.92)	+ (ICC=0.80)	NR	+ (r=0.20-0.42)	NR	NR	NR
COST-v1, *Australia version*	Durber et al., 2021	NR	+ (Cronbach's α=0.89)	+ (ICC=0.80)	NR	+ (r=-0.39-0.52)	NR	NR	NR
COST-v1, *Brazilian version*	de Alcantara Nogueira et al., 2020	- (RMSEA=1.20)	+ (Cronbach's α=0.83)	NR	NR	NR	- (p<0.01)	NR	NR
COST-v1, *India version*	Dar et al., 2021	- (EFA: χ^2= 60.82)	? (Cronbach's α=0.92)	NR	NR	NR	NR	NR	NR
COST-v1 *Simplified Chinese version*	Yu et al., 2021	- (CFI=0.86; SRMR=0.08)	+ (Cronbach's α=0.85-0.88)	+ (ICC=0.85)	NR	+ (r=-0.57-0.88)	NR	NR	? (no clear hypothesis)
COST-v2, *Italian version*	Ripamonti et al., 2020	? (CFI=1.00; RMSEA=0.04)	? (Cronbach's α=0.83)	+ (ICC=0.79)	NR	+ (r=-0.66-0.15)	NR	+ (β=-0.55)	NR
COST-v2, *Persian version*	Sharif et al., 2020	+ (CFI=0.97; RMSEA=0.07)	+ (Cronbach's α=0.89)	NR	NR	? (no clear hypothesis)	NR	NR	NR
COST-v2, *Traditional Chinese version*	Chan et al., 2021	- (CFI=0.91; RMSEA=0.15)	+ (Cronbach's α=0.86)	+ (ICC=0.71)	NR	+ (r=-0.46-0.47)	NR	NR	NR
CPILS employment/financial	Zhao et al., 2009	- (EFA: no model	+ (Cronbach's	NR	NR	? (no clear	NR	NR	NR

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domain		data)	α=0.78-0.97)			hypothesis)			
FIT	Hueniken et al., 2020	- (EFA: no model data)	+ (Cronbach's α=0.77)	+ (ICC=0.70)	NR	+ (r=0.26-0.61)	NR	NR	+ (r=-0.25)
IOC-CS financial problems domain	Zebrack et al., 2010	- (EFA: no model data)	- (Cronbach's α=0.70-0.86)	- (ICC≥0.75)	NR	- (>75% were not significant)	NR	? (no data)	NR
PFB	Veenstra et al., 2014	- (CFA: no model data)	+ (Cronbach's α=0.79)	NR	NR	NR	NR	NR	NR
PROFFIT	Riva et al., 2021	- (EFA: no model data)	+ (Cronbach's α=0.87)	+ (ICC=0.80)	NR	NR	NR	NR	NR
PRRS financial wellbeing domain	Shilling et al., 2018	- (EFA: no model data)	+ (Cronbach's α=0.90)	+ (ICC=0.86)	NR	+ (r=-0.71-0.65)	NR	+ (r=-0.65)	NR
P-SAFE, *French Version*	Tremblay et al., 2020	NR	NR	NR	NR	NR	? (no data)	NR	NR
QLACS financial problems domain	Avis et al., 2005	- (CFA: no model data)	+ (Cronbach's α=0.72)	NR	NR	+ (r=-0.57-0.60)	NR	- (r=-0.72)	NR
SDI-16 money matters domain	Wright et al., 2011	- (EFA: no model data)	+ (Cronbach's α=0.71-0.82)	NR	NR	NR	NR	NR	NR
SDI-21 providing for the family domain	Wright et al., 2005	- (EFA: no model data)	+ (Cronbach's α=0.50-0.86)	- (Weighted kappa=0.54-0.80)	NR	- (33% were not significant)	NR	? (r=-0.72)	NR
SFDQ	Dar et al., 2021	+ (CFI=0.98, TLI=0.97, RMSEA=0.045, SRMR=0.014)	+ (Cronbach's α=0.85-0.88)	NR	NR	NR	NR	NR	NR

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SWBS	Head, 2008; Head & Faul, 2008	- (CFA: no model data)	+ (Cronbach's α=0.92)	NR	NR	+ (r=-0.57-0.60)	NR	+ (r=-0.12-0.03)	NR

CCEQ: Chronic Cancer Experiences Questionnaire; CFA: Confirmatory Factor Analysis; CFI: Comparative Fit Index; COST: Comprehensive Score for Financial Toxicity; CPILS: Cancer Problems in Living Scale; EFA: Exploratory Factor Analysis; FIT: Financial Index of Toxicity; ICC: Intraclass Correlation Coefficient; IOC-CS: Impact of Cancer-Childhood Survivors; NA: Not Applicable; NR: Not Report; PFB: Personal Financial Burden ; PROM: Patient-Reported Outcome Measures; PROFFIT: Patient-Reported Outcome for Fighting Financial Toxicity; PRRS: Patient Roles and Responsibilities Scale ; P-SAFE: Patient Self-Administered Financial Effects questionnaire; QLACS: Quality of Life in Adult Cancer Survivors; RMSEA: Root Mean Square Error of Approximation; SDI: Social Difficulties Inventory Cancer Care Outcomes; SFDQ: Subjective Financial Distress Questionnaire; SWBS: Socioeconomic Wellbeing Scale

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Table 4 Certainty of evidence of measurement properties

PROM	Author (year)	Structural validity	Internal consistency	Reliability	Measurement error	Hypothesis testing	Cross-cultural validity	Criterion validity	Responsiveness
CCEQ financial advice domain	Harley et al., 2019	Moderate	High	-	-	High	-	-	-
COST-v1	De Souza et al., 2014; De Souza et al., 2017	-	High	Moderate	-	High	-	-	-
COST-v1, *Australia version*	Durber et al., 2021	-	High	Moderate	-	High	-	-	-
COST-v1, *Brazilian version*	de Alcantara Nogueira et al., 2020	High	High	-	-	-	Very low	-	-
COST-v1, *India version*	Dar et al., 2021	Very Low	Low	-	-	-	-	-	-
COST-v1, *Simplified Chinese version*	Yu et al., 2021	High	High	High	-	Low	-	-	Low
COST-v2, *Italian version*	Ripamonti et al., 2020	Very low	High	Very low	-	High	-	High	-
COST-v2, *Persian version*	Sharif et al., 2020	High	High	-	-	Very low	-	-	-
COST-v2, *Traditional Chinese version*	Chan et al., 2021	High	High	Low	-	High	-	-	-
CPILS employment/financial domain	Zhao et al., 2009	High	High	-	-	High	-	-	-
FIT	Hueniken et al., 2020	Low	Moderate	Very Low	-	Moderate	-	-	Moderate
IOC-CS financial problems domain	Zebrack et al., 2010	Moderate	High	Moderate	-	High	-	Very low	-
PFB	Veenstra et al., 2014	Moderate	Very low	-	-	-	-	-	-
PROFFIT	Riva et al., 2021	Moderate	High	Moderate	-	-	-	-	-

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PRRS financial wellbeing domain	Shilling et al., 2018	Moderate	High	Very low	-	Very low	-	Very low	-
P-SAFE, *French Version*	Tremblay et al., 2020	-	-	-	-	-	Very low	-	-
QLACS financial problems domain	Avis et al., 2005	Moderate	High	-	-	Low	-	Moderate	-
SDI-16 money matters domain	Wright et al., 2011	High	High	Very low	-	-	-	Very low	-
SDI-21 providing for the family domain	Wright et al., 2005	High	High	-	-	-	-	-	-
SFDQ	Dar et al., 2022	High	High	-	-	-	-	-	-
SWBS	Head, 2008; Head & Faul, 2008	High	High	-	-	High	-	-	-

CCEQ: Chronic Cancer Experiences Questionnaire; COST: Comprehensive Score for Financial Toxicity; CPILS: Cancer Problems in Living Scale; FIT: Financial Index of Toxicity; IOC-CS: Impact of Cancer-Childhood Survivors; PFB: Personal Financial Burden; PROM: Patient-Reported Outcome Measures; PROFFIT: Patient-Reported Outcome for Fighting Financial Toxicity; PRRS: Patient Roles and Responsibilities Scale ; P-SAFE: Patient Self-Administered Financial Effects questionnaire; QLACS: Quality of Life in Adult Cancer Survivors; RMSEA: Root Mean Square Error of Approximation; SDI: Social Difficulties Inventory Cancer Care Outcomes; SFDQ: Subjective Financial Distress Questionnaire; SWBS: Socioeconomic Wellbeing Scale

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Figure 1 PRISMA Flow Chat of Selection Process

Appendix I Search Strategy and Results

Appendix II COSMIN Risk of Bias Checklist

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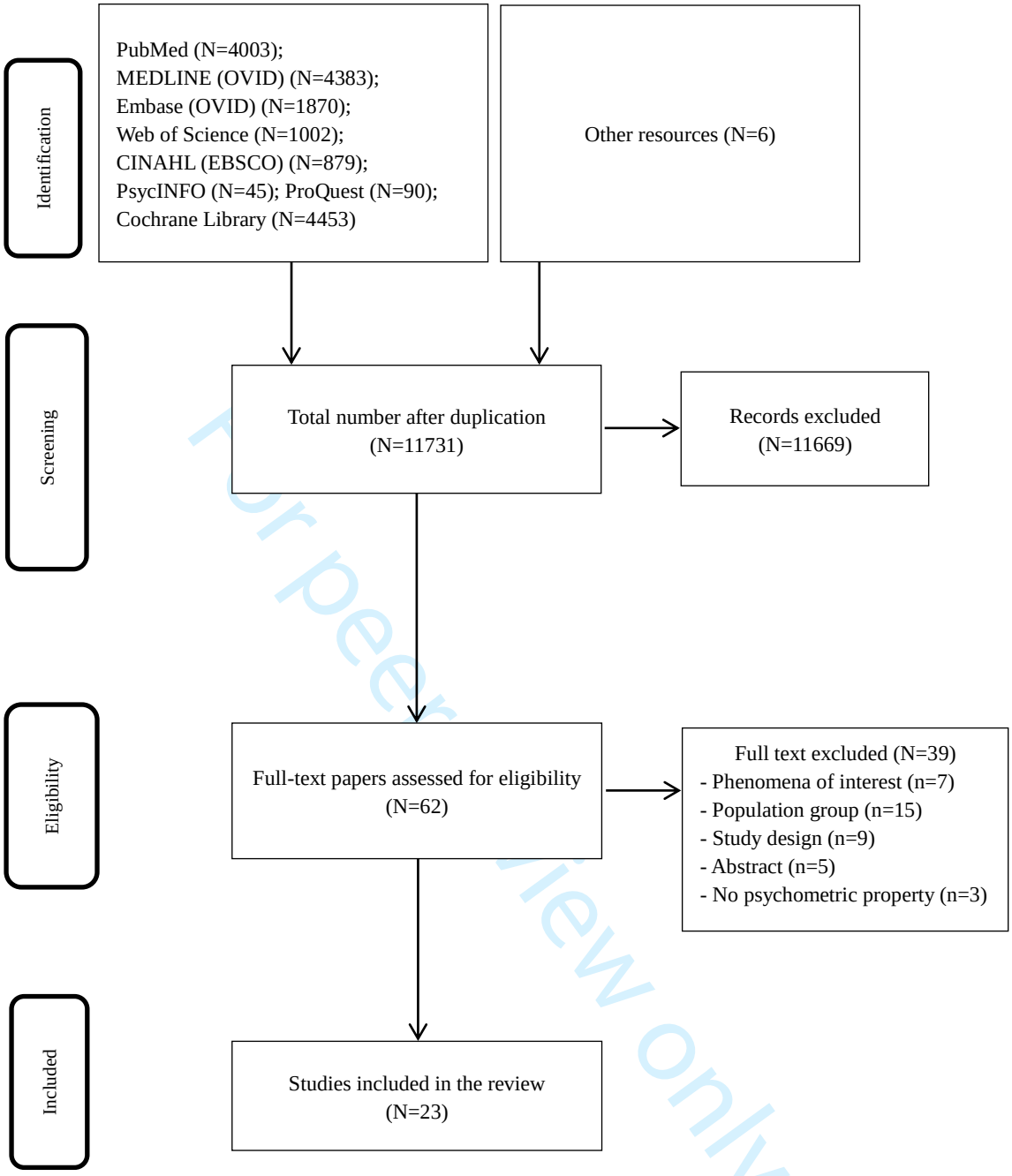


Figure 1 PRISMA flow chat of selection process

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Appendix I Search strategy and results

Search strategy for PubMed

Search time: 2022-3-13 15:07 (UTC+8)

Search	Query	Items found
#1	Cancer[Title/Abstract] OR neoplasms[MeSH]	4176348
#2	Patient?[Title/Abstract] OR survivor?[Title/Abstract] OR patients[MeSH] OR "cancer survivors"[MeSH] OR survivors[MeSH]	2721864
#3	Cost[Title/Abstract] OR bill?[Title/Abstract] OR expense[Title/Abstract] OR "productivity loss"[Title/Abstract] OR "out-of-pocket"[Title/Abstract] OR "economic burden"[Title/Abstract] OR "financial toxicity"[Title/Abstract] OR "financial hardship"[Title/Abstract] OR "financial burden"[Title/Abstract] OR "financial effect"[Title/Abstract] OR "financial stress"[Title/Abstract] OR "economic burden"[Title/Abstract] OR "economic hardship"[Title/Abstract] OR "co-payment"[Title/Abstract] OR "cost of illness"[MeSH]	573062
#4	Scale?[Title/Abstract] OR "patient reported outcome measur*"[Title/Abstract] OR PROM? [Title/Abstract] OR measure* [Title/Abstract] OR "Patient Reported Outcome Measures*"[MeSH] OR "Surveys and Questionnaires"[MeSH]	5065964
#5	(instrumentation[sh] OR methods[sh] OR "Validation Studies"[pt] OR "Comparative Study"[pt] OR "psychometrics"[MeSH] OR psychometr*[tiab] OR clinimetr*[tw] OR clinometr*[tw] OR "outcome assessment (health care)"[MeSH] OR "outcome assessment"[tiab] OR "outcome measure*"[tw] OR "observer variation"[MeSH] OR "observer variation"[tiab] OR "Health Status Indicators"[Mesh] OR "reproducibility of results"[MeSH] OR reproducib*[tiab] OR "discriminant analysis"[MeSH] OR reliab*[tiab] OR unreliab*[tiab] OR valid*[tiab] OR "coefficient of variation"[tiab] OR coefficient[tiab] OR homogeneity[tiab] OR homogeneous[tiab] OR "internal consistency"[tiab] OR (cronbach*[tiab] AND (alpha[tiab] OR alphas[tiab])) OR (item[tiab] AND (correlation*[tiab] OR selection*[tiab] OR reduction*[tiab])) OR agreement[tw] OR precision[tw] OR imprecision[tw] OR "precise values"[tw] OR test-retest[tiab] OR (test[tiab] AND retest[tiab]) OR (reliab*[tiab] AND (test[tiab] OR retest[tiab])) OR stability[tiab] OR interrater[tiab] OR inter-rater[tiab] OR intrarater[tiab] OR intra-rater[tiab] OR intertester[tiab] OR inter-tester[tiab] OR intratester[tiab] OR intra-tester[tiab] OR interobserver[tiab] OR inter-observer[tiab] OR intraobserver[tiab] OR intra-observer[tiab] OR intertechnician[tiab] OR inter-technician[tiab] OR intratechnician[tiab] OR intra-technician[tiab] OR interexaminer[tiab] OR inter-examiner[tiab] OR intraexaminer[tiab] OR intra-examiner[tiab] OR interassay[tiab] OR inter-assay[tiab] OR intraassay[tiab] OR intra-assay[tiab] OR interindividual[tiab] OR inter-individual[tiab] OR intraindividual[tiab] OR intra-individual[tiab] OR interparticipant[tiab] OR inter-participant[tiab] OR intraparticipant[tiab] OR intra-participant[tiab] OR kappa[tiab] OR kappa's[tiab] OR kappas[tiab] OR repeatab*[tw] OR ((replicab*[tw] OR repeated[tw]) AND (measure[tw] OR measures[tw] OR findings[tw] OR result[tw] OR results[tw] OR test[tw] OR tests[tw])) OR generaliza*[tiab] OR generalisa*[tiab] OR concordance[tiab] OR (intraclass[tiab] AND correlation*[tiab]) OR discriminative[tiab] OR "known group"[tiab] OR "factor analysis"[tiab] OR "factor analyses"[tiab] OR "factor structure"[tiab] OR "factor structures"[tiab] OR dimension*[tiab] OR subscale*[tiab] OR (multitrait[tiab] AND scaling[tiab] AND (analysis[tiab] OR analyses[tiab])) OR "item discriminant"[tiab] OR "interscale correlation*"[tiab] OR error[tiab] OR errors[tiab] OR "individual variability"[tiab]	10114928

	OR “interval variability”[tiab] OR “rate variability”[tiab] OR (variability[tiab] AND (analysis[tiab] OR values[tiab])) OR (uncertainty[tiab] AND (measurement[tiab] OR measuring[tiab])) OR “standard error of measurement”[tiab] OR sensitiv*[tiab] OR responsive*[tiab] OR (limit[tiab] AND detection[tiab]) OR “minimal detectable concentration”[tiab] OR interpretab*[tiab] OR ((minimal[tiab] OR minimally[tiab] OR clinical[tiab] OR clinically[tiab]) AND (important[tiab] OR significant[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR (small*[tiab] AND (real[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR “meaningful change”[tiab] OR “ceiling effect”[tiab] OR “floor effect”[tiab] OR “Item response model”[tiab] OR IRT[tiab] OR Rasch[tiab] OR “Differential item functioning”[tiab] OR DIF[tiab] OR “computer adaptive testing”[tiab] OR “item bank”[tiab] OR “cross-cultural equivalence”[tiab])	
#6	(“addresses”[Publication Type] OR “biography”[Publication Type] OR “case reports”[Publication Type] OR “comment”[Publication Type] OR “directory”[Publication Type] OR “editorial”[Publication Type] OR “festschrift”[Publication Type] OR “interview”[Publication Type] OR “lectures”[Publication Type] OR “legal cases”[Publication Type] OR “legislation”[Publication Type] OR “letter”[Publication Type] OR “news”[Publication Type] OR “newspaper article”[Publication Type] OR “patient education handout”[Publication Type] OR “popular works”[Publication Type] OR “congresses”[Publication Type] OR “consensus development conference”[Publication Type] OR “consensus development conference, nih”[Publication Type] OR “practice guideline”[Publication Type]) NOT (“animals”[MeSH Terms] NOT “humans”[MeSH Terms])	4415236
#7	#1 AND #2 AND #3 AND #4 AND #5	4097
#8	#7 NOT #6	4003

Search strategy for MEDLINE (Ovid)

Search time: 2022-3-13 15:12 (UTC+8)

Search	Query	Items found
#1	cancer.ab. or cancer.ti. or neoplasms.hw.	3208957
#2	patient?.ab. or patient?.ti. or survivor?.ab. or survivor?.ti. or patients.hw. or cancer survivors.hw. or survivors.hw.	6631761
#3	Cost.ab. or Cost.ti. or bill?.ab. or bill?.ti. or expense.ab. or expense.ti. or productivity loss.ab. or productivity loss.ti. or out-of-pocket.ab. or out-of-pocket.ti. or economic burden.ab. or economic burden.ti. or financial toxicity.ab. or financial toxicity.ti. or financial hardship.ab. or financial hardship.ti. or financial burden.ab. or financial burden.ti. or financial effect.ab. or financial effect.ti. or financial stress.ab. or financial stress.ti. or economic burden.ab. or economic burden.ti. or economic hardship.ab. or economic hardship.ti. or co-payment.ab. or co-payment.ti. or (cost of illness).hw.	459262
#4	Scale?.ab. or Scale?.ti. or patient reported outcome measur*.ab. or patient reported outcome measur*.ti. or PROM?.ab. or PROM?.ti. or measure*.ab. or measure*.ti. or Patient Reported Outcome Measures*.ab. or Patient Reported Outcome Measures*.ti. or (Surveys and Questionnaires).hw.	3951480
#5	(instrumentation or methods).fs.	4440288
#6	(Validation Studies or Comparative Study).pt.	1910291

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7	exp Psychometrics/	83581
8	psychometr*.ti,ab.	46299
9	(clinimetr* or clinometr*).tw.	1135
10	exp Outcome Assessment Health Care/	1270817
11	outcome assessment.ti,ab.	3870
12	outcome measure*.tw.	230270
13	exp Observer Variation/	44491
14	observer variation.ti,ab.	1055
15	exp Health Status Indicators/	335480
16	exp Reproducibility of Results/	442637
17	reproducib*.ti,ab.	152648
18	exp Discriminant Analysis/	11624
19	(reliab* or unreliab* or valid* or coefficient or homogeneity or homogeneous or internal consistency).ti,ab.	1334500
20	(cronbach* and (alpha or alphas)).ti,ab.	22858
21	(item and (correlation* or selection* or reduction*)).ti,ab.	22439
22	(agreement or precision or imprecision or precise values or test-retest).ti,ab.	369056
23	(test and retest).ti,ab.	26981
24	(reliab* and (test or retest)).ti,ab.	87641
25	(stability or interrater or inter-rater or intrarater or intra-rater or intertester or inter-tester or intratester or intra-tester or interobserver or inter-observer or intraobserver or intraobserver or intertechnician or inter-technician or intratechnician or intra-technician or interexaminer or inter-examiner or intraexaminer or intra-examiner or interassay or interassay or intraassay or intra-assay or interindividual or inter-individual or intraindividual or intra-individual or interparticipant or inter-participant or intraparticipant or intra-participant or kappa or kappa's or kappas or repeatab*).ti,ab.	551861
26	((replicab* or repeated) and (measure or measures or findings or result or results or test or tests)).ti,ab.	192657
27	(generaliza* or generalisa* or concordance).ti,ab.	87243
28	(intraclass and correlation*).ti,ab.	26027
29	(discriminative or known group or factor analysis or factor analyses or dimension* or subscale*).ti,ab.	547916
30	(multitrait and scaling and (analysis or analyses)).ti,ab.	134
31	(item discriminant or interscale correlation* or error or errors or individual variability).ti,ab.	273771
32	(variability and (analysis or values)).ti,ab.	97699
33	(uncertainty and (measurement or measuring)).ti,ab.	5737
34	(standard error of measurement or sensitiv* or responsive*).ti,ab.	1522843
35	((minimal or minimally or clinical or clinically) and (important or significant or detectable) and (change or difference)).ti,ab.	236237
36	(small* and (real or detectable) and (change or difference)).ti,ab.	7151
37	(meaningful change or ceiling effect or floor effect or Item response model or IRT or Rasch or Differential item functioning or DIF or computer adaptive testing or item bank or cross-cultural equivalence).ti,ab.	13325

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38	5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37	9593937
39	(child* or pediatric* or infan* or neonat* or newborn* or teen* or youth*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept, rare disease supplementary concept, unique identifier]	3289302
40	1 and 2 and 3 and 4 and 38	4635
41	40 not 39	4383

Search strategy for EMBASE (Ovid)

Search time: 2022-3-13 16:17 (UTC+8)

Search	Query	Items found
#1	cancer.ab. or cancer.ti. or neoplasms.hw.	2728162
#2	patient?.ab. or patient?.ti. or survivor?.ab. or survivor?.ti. or patients.hw. or cancer survivors.hw. or survivors.hw.	10910535
#3	Cost.ab. or Cost.ti. or bill?.ab. or bill?.ti. or expense.ab. or expense.ti. or productivity loss.ab. or productivity loss.ti. or out-of-pocket.ab. or out-of-pocket.ti. or economic burden.ab. or economic burden.ti. or financial toxicity.ab. or financial toxicity.ti. or financial hardship.ab. or financial hardship.ti. or financial burden.ab. or financial burden.ti. or financial effect.ab. or financial effect.ti. or financial stress.ab. or financial stress.ti. or economic burden.ab. or economic burden.ti. or economic hardship.ab. or economic hardship.ti. or co-payment.ab. or co-payment.ti. or (cost of illness).hw.	743789
#4	Scale?.ab. or Scale?.ti. or patient reported outcome measur*.ab. or patient reported outcome measur*.ti. or PROM?.ab. or PROM?.ti. or measure*.ab. or measure*.ti. or Patient Reported Outcome Measures*.ab. or Patient Reported Outcome Measures*.ti. or (Surveys and Questionnaires).hw.	5577750
#5	exp Intermethod comparison/ OR exp data collection method/ OR exp validation study/ OR exp feasibility study/ OR exp pilot study/ OR exp psychometry/ OR exp reproducibility/ OR reproducib*:ab,ti OR audit:ab,ti OR psychometr*:ab,ti OR clinimetr*:ab,ti OR clinometr*:ab,ti OR exp observer variation/ OR observer variation:ab,ti OR exp discriminant analysis/ OR exp validity/ OR reliab*:ab,ti OR valid*:ab,ti OR coefficient:ab,ti OR internal consistency:ab,ti OR (cronbach*:ab,ti AND (alpha:ab,ti OR alphas:ab,ti)) OR item correlation:ab,ti OR item correlations:ab,ti OR item selection:ab,ti OR item selections:ab,ti OR item reduction:ab,ti OR item reductions:ab,ti OR agreement:ab,ti OR precision:ab,ti OR imprecision:ab,ti OR precise values:ab,ti OR test-retest:ab,ti OR (test:ab,ti AND retest:ab,ti) OR (reliab*:ab,ti AND (test:ab,ti OR retest:ab,ti)) OR stability:ab,ti OR interrater:ab,ti OR inter-rater:ab,ti OR intrarater:ab,ti OR intra-rater:ab,ti OR intertester:ab,ti OR inter-tester:ab,ti OR intratester:ab,ti OR intratester:ab,ti OR interobeserver:ab,ti OR inter-observer:ab,ti OR intraobserver:ab,ti OR intraobserver:ab,ti OR intertechnician:ab,ti OR inter-technician:ab,ti OR intratechnician:ab,ti OR intratechnician:ab,ti OR interexaminer:ab,ti OR inter-examiner:ab,ti OR intraexaminer:ab,ti OR intraexaminer:ab,ti OR interassay:ab,ti OR inter-assay:ab,ti OR intraassay:ab,ti OR intra-assay:ab,ti OR interindividual:ab,ti OR inter-individual:ab,ti OR intraindividual:ab,ti OR intra-individual:ab,ti OR interparticipant:ab,ti OR inter-participant:ab,ti OR intraparticipant:ab,ti OR intraparticipant:ab,ti OR kappa:ab,ti OR kappas:ab,ti OR coefficient of variation:ab,ti OR	2229098

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	repeatab*:ab,ti OR (replicab*:ab,ti OR repeated:ab,ti AND (measure:ab,ti OR measures:ab,ti OR findings:ab,ti OR result:ab,ti OR results:ab,ti OR test:ab,ti OR tests:ab,ti)) OR generaliza*:ab,ti OR generalisa*:ab,ti OR concordance:ab,ti OR (intraclass:ab,ti AND correlation*:ab,ti) OR discriminative:ab,ti OR known group:ab,ti OR factor analysis:ab,ti OR factor analyses:ab,ti OR factor structure:ab,ti OR factor structures:ab,ti OR dimensionality:ab,ti OR subscale*:ab,ti OR multitrait scaling analysis:ab,ti OR multitrait scaling analyses:ab,ti OR item discriminant:ab,ti OR interscale correlation:ab,ti OR interscale correlations:ab,ti OR (error:ab,ti OR errors:ab,ti AND (measure*:ab,ti OR correlat*:ab,ti OR evaluat*:ab,ti OR accuracy:ab,ti OR accurate:ab,ti OR precision:ab,ti OR mean:ab,ti)) OR individual variability:ab,ti OR interval variability:ab,ti OR rate variability:ab,ti OR variability analysis:ab,ti OR (uncertainty:ab,ti AND (measurement:ab,ti OR measuring:ab,ti)) OR standard error of measurement:ab,ti OR sensitiv*:ab,ti OR responsive*:ab,ti OR (limit:ab,ti AND detection:ab,ti) OR minimal detectable concentration:ab,ti OR interpretab*:ab,ti OR (small*:ab,ti AND (real:ab,ti OR detectable:ab,ti) AND (change:ab,ti OR difference:ab,ti)) OR meaningful change:ab,ti OR minimal important change:ab,ti OR minimal important difference:ab,ti OR minimally important change:ab,ti OR minimally important difference:ab,ti OR minimal detectable change:ab,ti OR minimal detectable difference:ab,ti OR minimally detectable change:ab,ti OR minimally detectable difference:ab,ti OR minimal real change:ab,ti OR minimal real difference:ab,ti OR minimally real change:ab,ti OR minimally real difference:ab,ti OR ceiling effect:ab,ti OR floor effect:ab,ti OR item response model:ab,ti OR irt:ab,ti OR rasch:ab,ti OR differential item functioning:ab,ti OR dif:ab,ti OR computer adaptive testing:ab,ti OR item bank:ab,ti OR cross-cultural equivalence:ab,ti	
#6	1 and 2 and 3 and 4 and 5	1870

Search strategy for CINAHL (EBSCO)

Search time: 2022-3-13 17:10 (UTC+8)

Search	Query	Items found
#1	TI cancer OR AB cancer OR MH neoplasms	478227
#2	AB patient? OR TI patient? OR AB survivor? OR TI survivor? OR MH patients OR MH “cancer survivors” OR MH survivors	2004224
#3	AB Cost OR TI Cost OR AB bill? OR TI bill? OR AB expense OR TI expense OR AB “productivity loss” OR TI “productivity loss” OR AB out-of-pocket OR TI out-of-pocket OR AB “economic burden” OR TI “economic burden” OR AB “financial toxicity” OR TI “financial toxicity” OR AB “financial hardship” OR TI “financial hardship” OR AB “financial burden” OR TI “financial burden” OR AB “financial effect” OR TI “financial effect” OR AB “financial stress” OR TI “financial stress” OR AB “economic burden” OR TI “economic burden” OR AB “economic hardship” OR TI “economic hardship” OR AB co-payment OR TI co-payment OR MH “cost of illness”	211023
#4	AB Scale? OR TI Scale? OR AB “patient reported outcome measur*” OR TI “patient reported outcome measur*” OR AB PROM? OR TI PROM? OR AB measure* OR TI measure* OR AB “Patient Reported Outcome Measures*” OR MH “Patient Reported Outcome Measures*” OR MH “Surveys and Questionnaires”	942889
#5	(MH “Psychometrics”) or (TI psychometr* or AB psychometr*) or (TI clinimetr* or AB	702880

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	clinimetr*) or (TI clinometr* OR AB clinometr*) or (MH "Outcome Assessment") or (TI outcome assessment or AB outcome assessment) or (TI outcome measure* or AB outcome measure*) or (MH "Health Status Indicators") or (MH "Reproducibility of Results") or (MH"Discriminant Analysis") or ((TI reproducib* or AB reproducib*) or (TI reliab* or AB reliab*) or (TI unreliab* or AB unreliab*)) or ((TI valid* or AB valid*) or (TI coefficient or AB coefficient) or (TI homogeneity or AB homogeneity)) or (TI homogeneous or AB homogeneous) or (TI "coefficient of variation" or AB "coefficient of variation") or (TI "internal consistency" or AB "internal consistency") or (MH "Internal Consistency+") or (MH "Reliability+") or (MH "Measurement Error+") or (MH "Content Validity+") or "hypothesis testing" or "structural validity" or "cross-cultural validity" or (MH "Criterion-Related Validity+") or "responsiveness" or "interpretability" or (TI reliab* or AB reliab*) and ((TI test or AB test) OR (TI retest or AB retest)) or (TI stability or AB stability) or (TI interrater or AB interrater) or (TI inter-rater or AB inter-rater) or (TI intrarater or AB intrarater) or (TI intra-rater or AB intrarater) or (TI intertester or AB intertester) or (TI inter-tester or AB inter-tester) or (TI intratester or AB intratester) or (TI intra-tester or AB intra-tester) or (TI interobserver or AB interobserver) or (TI inter-observer or AB inter-observer) or (TI intraobserver or AB intraobserver) or (TI intra-observer or AB intra-observer) or (TI intertechnician or AB intertechnician) or (TI inter-technician or AB inter-technician) or (TI intratechnician or AB intratechnician) or (TI intra-technician or AB intra-technician) or (TI interexaminer or AB interexaminer) or (TI inter-examiner or AB inter-examiner) or (TI intraexaminer or AB intraexaminer) OR (TI intra-examiner or AB intra-examiner) or (TI intra-examiner or AB intraexaminer) or (TI interassay or AB interassay) or (TI inter-assay or AB inter-assay) or (TI intraassay or AB intraassay) or (TI intra-assay or AB intra-assay) or (TI interindividual or AB interindividual) or (TI inter-individual or AB inter-individual) OR (TI intraindividual or AB intraindividual) or (TI intra-individual or AB intra-individual) or (TI interparticipant or AB interparticipant) or (TI inter-participant or AB inter-participant) or (TI intraparticipant or AB intraparticipant) or (TI intra-participant or AB intra-participant) or (TI kappa or AB kappa) or (TI kappa's or AB kappa's) or (TI kappas or AB kappas) or (TI repeatab* or AB repeatab*) or (TI responsive* or AB responsive*) or (TI interpretab* or AB interpretab*)	
#6	1 and 2 and 3 and 4 and 5	879

Search strategy for PsycINFO (EBSCO)

Search time: 2022-3-13 17:32 (UTC+8)

Search	Query	Items found
#1	TI cancer OR AB cancer OR MH neoplasms	1240
#2	AB patient? OR TI patient? OR AB survivor? OR TI survivor? OR MH patients OR MH "cancer survivors" OR MH survivors	17860
#3	AB Cost OR TI Cost OR AB bill? OR TI bill? OR AB expense OR TI expense OR AB "productivity loss" OR TI "productivity loss" OR AB out-of-pocket OR TI out-of-pocket OR AB "economic burden" OR TI "economic burden" OR AB "financial toxicity" OR TI "financial toxicity" OR AB "financial hardship" OR TI "financial hardship" OR AB "financial burden" OR TI "financial burden" OR AB "financial effect" OR TI "financial effect" OR AB "financial	4217

	stress" OR TI "financial stress" OR AB "economic burden" OR TI "economic burden" OR AB "economic hardship" OR TI "economic hardship" OR AB co-payment OR TI co-payment OR MH "cost of illness"	
#4	AB Scale? OR TI Scale? OR AB "patient reported outcome measur*" OR TI "patient reported outcome measur*" OR AB PROM? OR TI PROM? OR AB measure* OR TI measure* OR AB "Patient Reported Outcome Measures*" OR MH "Patient Reported Outcome Measures*" OR MH "Surveys and Questionnaires"	50879
#6	1 and 2 and 3 and 4	45

Search strategy for Web of Science

Search time: 2022-3-13 18:17 (UTC+8)

Search	Query	Items found
#1	TI=cancer OR TS= (cancer OR neoplasms)	6889664
#2	TI=(patient? OR survivor?) OR TS=(patients OR "cancer survivors" OR survivors)	12513517
#3	TI=(Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress" OR "economic burden" OR "economic hardship") OR TS="cost of illness"	463395
#4	TI=(Scale? OR "patient reported outcome measur*" OR PROM? OR measure*) OR TS=("Patient Reported Outcome Measures*" OR "Surveys and Questionnaires")	2955804
#6	#4 AND #3 AND #2 AND #1	1002

Search strategy for ProQuest Dissertations and Theses

Search time: 2022-3-13 18:41 (UTC+8)

Search	Query	Items found
#1	ti(cancer OR neoplasms) OR su(cancer OR neoplasms)	48268
#2	ti(patient? OR survivor? OR "cancer survivors") OR su(patient? OR survivor? OR cancer survivors)	51934
#3	ti(Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress") OR su(Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress") OR ti("economic burden" OR "economic hardship" OR "cost of illness") OR su("economic burden" OR "economic hardship" OR "cost of illness")	45357
#4	All : Scale? OR "patient reported outcome measur*" OR PROM? OR measure*	2854129
#6	#4 AND #3 AND #2 AND #1	90

Search strategy for Cochrane Library (Wiley)

Search time: 2022-3-13 19:05 (UTC+8)

Search	Query	Items found

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#1	(Cancer):ti,ab,kw	177850
#2	MeSH descriptor: [Neoplasms] explode all trees	86823
#3	(patient? OR survivor? OR "cancer survivors"):ti,ab,kw	1070252
#4	MeSH descriptor: [Patients] explode all trees	2946
#5	MeSH descriptor: [Cancer Survivors] explode all trees	476
#6	MeSH descriptor: [Survivors] explode all trees	1760
#7	(Cost OR bill? OR expense OR "productivity loss" OR out-of-pocket OR "economic burden" OR "financial toxicity" OR "financial hardship" OR "financial burden" OR "financial effect" OR "financial stress" OR "economic burden" OR "economic hardship" OR "cost of illness"):ti,ab,kw	67224
#8	MeSH descriptor: [Cost of Illness] explode all trees	864
#9	(Scale? OR "patient reported outcome measur*" OR PROM? OR measure*):ti,ab,kw	602173
#10	MeSH descriptor: [Patient Reported Outcome Measures] explode all trees	929
#11	MeSH descriptor: [Surveys and Questionnaires] explode all trees	58363
#12	("Validation Studies" OR "Comparative Study"):pt	169332
#13	MeSH descriptor: [Psychometrics] explode all trees	2951
#14	(psychometr*):ti,ab,kw	6755
#15	#1 OR #2	208929
#16	#3 OR #4 OR #5 OR #6	1070725
#17	#7 OR #8 OR #9 OR #10 OR #11	666816
#18	#12 OR #13 OR #14	175091
#19	#15 AND #16 AND #17 AND #18	4453

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Appendix II COSMIN Checklists

eTable 1 COSMIN risk of bias checklist

PROM Development	Results
1. Is a clear description provided of the construct to be measured?	
2. Is the origin of the construct clear: was a theory, conceptual framework or disease model used or clear rationale provided to define the construct to be measured?	
3. Is a clear description provided of the target population for which the PROM was developed?	
4. Is a clear description provided of the context of use	
5. Was the PROM development study performed in a sample representing the target population for which the PROM was developed?	
6. Was an appropriate qualitative data collection method used to identify relevant items for a new PROM?	
7. Were skilled group moderators/interviewers used?	
8. Were the group meetings or interviews based on an appropriate topic or interview guide?	
9. Were the group meetings or interviews recorded and transcribed verbatim?	
10. Was an appropriate approach used to analyse the data?	
11. Was at least part of the data coded independently?	
12. Was data collection continued until saturation was reached?	
13. For quantitative studies (surveys): was the sample size appropriate?	
14. Was a cognitive interview study or other pilot test conducted?	
15. Was the cognitive interview study or other pilot test performed in a sample representing the target population?	
16. Were patients asked about the comprehensibility of the PROM?	
17. Were all items tested in their final form?	
18. Was an appropriate qualitative method used to assess the comprehensibility of the PROM instructions, items, response options, and recall period?	
19. Was each item tested in an appropriate number of patients?	
20. Were skilled interviewers used?	
21. Were the interviews based on an appropriate interview guide?	
22. Were the interviews recorded and transcribed verbatim?	
23. Was an appropriate approach used to analyse the data?	
24. Were at least two researchers involved in the analysis?	
25. Were problems regarding the comprehensibility of the PROM instructions, items, response options, and recall period appropriately addressed by adapting the PROM?	
26. Were patients asked about the comprehensiveness of the PROM?	
27. Was the final set of items tested?	
28. Was an appropriate method used for assessing the comprehensiveness of the PROM?	
29. Was each item tested in an appropriate number of patients?	
30. Were skilled interviewers used?	
31. Were the interviews based on an appropriate interview guide?	
32. Were the interviews recorded and transcribed verbatim?	

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33. Was an appropriate approach used to analyse the data?	
34. Were at least two researchers involved in the analysis?	
35. Were problems regarding the comprehensiveness of the PROM appropriately addressed by adapting the PROM?	
Content validity	
1. Was an appropriate method used to ask patients whether each item is relevant for their experience with the condition?	
2. Was each item tested in an appropriate number of patients?	
3. Were skilled group moderators/interviewers used?	
4. Were the group meetings or interviews based on an appropriate topic or interview guide?	
5. Were the group meetings or interviews recorded and transcribed verbatim?	
6. Was an appropriate approach used to analyse the data?	
7. Were at least two researchers involved in the analysis?	
8. Was an appropriate method used for assessing the comprehensiveness of the PROM?	
9. Was each item tested in an appropriate number of patients?	
10. Were skilled group moderators/interviewers used?	
11. Were the group meetings or interviews based on an appropriate topic or interview guide?	
12. Were the group meetings or interviews recorded and transcribed verbatim?	
13. Was an appropriate approach used to analyse the data?	
14. Were at least two researchers involved in the analysis?	
15. Was an appropriate qualitative method used for assessing the comprehensibility of the PROM instructions, items, response options, and recall period?	
16. Was each item tested in an appropriate number of patients?	
17. Were skilled group moderators/interviewers used?	
18. Were the group meetings or interviews based on an appropriate topic or interview guide?	
19. Were the group meetings or interviews recorded and transcribed verbatim?	
20. Was an appropriate approach used to analyse the data?	
21. Were at least two researchers involved in the analysis?	
22. Was an appropriate method used to ask professionals whether each item is relevant for the construct of interest?	
23. Were professionals from all relevant disciplines included?	
24. Was each item tested in an appropriate number of professionals?	
25. Was an appropriate approach used to analyse the data?	
26. Were at least two researchers involved in the analysis?	
27. Was an appropriate method used for assessing the comprehensiveness of the PROM?	
28. Were professionals from all relevant disciplines included?	
29. Was each item tested in an appropriate number of professionals?	
30. Was an appropriate approach used to analyse the data?	
31. Were at least two researchers involved in the analysis?	
Structural validity	

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1. For CTT: Was exploratory or confirmatory factor analysis performed?	
2. For IRT/Rasch: does the chosen model fit to the research question?	
3. Was the sample size included in the analysis adequate?	
4. Were there any other important flaws in the design or statistical methods of the study?	
Internal consistency	
1. Was an internal consistency statistic calculated for each unidimensional scale or subscale separately?	
2. For continuous scores: Was Cronbach's alpha or omega calculated?	
3. For dichotomous scores: Was Cronbach's alpha or KR-20 calculated?	
4. For IRT-based scores: Was standard error of the theta (SE (θ)) or reliability coefficient of estimated latent trait value (index of (subject or item) separation) calculated?	
5. Were there any other important flaws in the design or statistical methods of the study?	
Cross-cultural validity	
1. Were the samples similar for relevant characteristics except for the group variable?	
2. Was an appropriate approach used to analyse the data?	
3. Was the sample size included in the analysis adequate?	
4. Were there any other important flaws in the design or statistical methods of the study?	
Reliability	
1. Were patients stable in the interim period on the construct to be measured?	
2. Was the time interval appropriate?	
3. Were the test conditions similar for the measurements? e.g. type of administration, environment, instructions	
4. For continuous scores: Was an intraclass correlation coefficient (ICC) calculated?	
5. For dichotomous/nominal/ordinal scores: Was kappa calculated?	
6. For ordinal scores: Was a weighted kappa calculated?	
7. For ordinal scores: Was the weighting scheme described? e.g. linear, quadratic	
8. Were there any other important flaws in the design or statistical methods of the study?	
Measurement error	
1. Were patients stable in the interim period on the construct to be measured?	
2. Was the time interval appropriate?	
3. Were the test conditions similar for the measurements? (e.g. type of administration, environment, instructions)	
4. For continuous scores: Was the Standard Error of Measurement (SEM), Smallest Detectable Change (SDC) or Limits of Agreement (LoA) calculated?	
5. For dichotomous/nominal/ordinal scores: Was the percentage (positive and negative) agreement calculated?	
6. Were there any other important flaws in the design or statistical methods of the study?	
Criterion validity	
1. For continuous scores: Were correlations, or the area under the receiver operating curve calculated?	
2. For dichotomous scores: Were sensitivity and specificity determined?	
3. Were there any other important flaws in the design or statistical methods of the study?	
Hypotheses testing for construct validity	

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1. Is it clear what the comparator instrument(s) measure(s)?	
2. Were the measurement properties of the comparator instrument(s) sufficient?	
3. Was the statistical method appropriate for the hypotheses to be tested?	
4. Were there any other important flaws in the design or statistical methods of the study?	
5. Was an adequate description provided of important characteristics of the subgroups?	
6. Was the statistical method appropriate for the hypotheses to be tested?	
7. Were there any other important flaws in the design or statistical methods of the study?	
Responsiveness	
1. For continuous scores: Were correlations between change scores, or the area under the Receiver Operator Curve (ROC) curve calculated?	
2. For dichotomous scales: Were sensitivity and specificity (changed versus not changed) determined?	
3. Were there any other important flaws in the design or statistical methods of the study?	
4. Is it clear what the comparator instrument(s) measure(s)?	
5. Were the measurement properties of the comparator instrument(s) sufficient?	
6. Was the statistical method appropriate for the hypotheses to be tested?	
7. Were there any other important flaws in the design or statistical methods of the study?	
8. Was an adequate description provided of important characteristics of the subgroups?	
9. Was the statistical method appropriate for the hypotheses to be tested?	
10. Were there any other important flaws in the design or statistical methods of the study?	
11. Was an adequate description provided of the intervention given?	
12. Was the statistical method appropriate for the hypotheses to be tested?	
13. Were there any other important flaws in the design or statistical methods of the study?	

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eTable 2 Criteria for good measurement properties

Measurement property	Rating	Criteria
Structural validity	+	**CTT:** CFA: CFI or TLI or comparable measure >0.95 OR RMSEA <0.06 OR SRMR <0.082 **IRT/Rasch:** No violation of unidimensionality3: CFI or TLI or comparable measure >0.95 OR RMSEA <0.06 OR SRMR <0.08 AND no violation of local independence: residual correlations among the items after controlling for the dominant factor <0.20 OR Q3's < 0.37 AND no violation of monotonicity: adequate looking graphs OR item scalability >0.30 AND adequate model fit: IRT: $\chi 2$ >0.01 Rasch: infit and outfit mean squares ≥ 0.5 and ≤ 1.5 OR Z standardized values >-2 and <2
	?	CTT: Not all information for '+' reported IRT/Rasch: Model fit not reported
	-	Criteria for '+' not met
Internal consistency	+	At least low evidence for sufficient structural validity AND Cronbach's alpha(s) ≥ 0.70 for each unidimensional scale or subscale
	?	Criteria for "At least low evidence4 for sufficient structural validity" not met
	-	At least low evidence for sufficient structural validity AND Cronbach's alpha(s) < 0.70 for each unidimensional scale or subscale
Reliability	+	ICC or weighted Kappa ≥ 0.70
	?	ICC or weighted Kappa not reported
	-	ICC or weighted Kappa < 0.70
Measurement error	+	SDC or LoA < MIC
	?	MIC not defined
	-	SDC or LoA > MIC
Hypotheses testing for construct validity	+	The result is in accordance with the hypothesis
	?	No hypothesis defined (by the review team)
	-	The result is not in accordance with the hypothesis
Cross-cultural	+	No important differences found between group factors (such

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validity\measurement invariance		as age, gender, language) in multiple group factor analysis OR no important DIF for group factors (McFadden's R2 < 0.02)
	?	No multiple group factor analysis OR DIF analysis performed
	-	Important differences between group factors OR DIF was found
Criterion validity	+	Correlation with gold standard ≥ 0.70 OR AUC ≥ 0.70
	?	Not all information for '+' reported
	-	Correlation with gold standard < 0.70 OR AUC < 0.70
Responsiveness	+	The result is in accordance with the hypothesis7 OR AUC ≥ 0.70
	?	No hypothesis defined (by the review team)
	-	The result is not in accordance with the hypothesis7 OR AUC < 0.70

AUC: area under the curve; CFA: confirmatory factor analysis; CFI: comparative fit index; CTT: classical test theory; DIF: differential item functioning; ICC: intraclass correlation coefficient; IRT: item response theory; LoA: limits of agreement; MIC: minimal important change; RMSEA: Root Mean Square Error of Approximation; SEM: Standard Error of Measurement; SDC: smallest detectable change; SRMR: Standardized Root Mean Residuals; TLI: Tucker-Lewis index; "+": sufficient; "-": insufficient; "?": indeterminate.

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eTable 3 Modified GRADE approach for assessing certainty of evidence*

Domain	Grade	Reason
Risk of bias	-0 level: No	There are multiple studies of at least adequate quality, or there is one study of very good quality available
	-1 level: Serious	There are multiple studies of doubtful quality available, or there is only one study of adequate quality
	-2 level: Very serious	There are multiple studies of inadequate quality, or there is only one study of doubtful quality available
	-3 level: Extremely serious	There is only one study of inadequate quality available
Inconsistency	-0 level: No	There is no inconsistency among pooled studies or there is only one study in subgroups
	-1 level: Serious	There are severe inconsistencies among pooled studies
	-2 level: Very serious	There are very severe inconsistencies among pooled studies.
Imprecision	-0 level: No	Total sample size>50-100
	-1 level: Serious	Total sample size=50-100
	-2 level: Very serious	Total sample size <50
Indirectness	-0 level: No	There is no indirectness between results and conclusion
	-1 level: Serious	There is severe indirectness between results and conclusion
	-2 level: Very serious	There is very severe indirectness between results and conclusion

*The starting point of quality level is high evidence. The quality of evidence is subsequently downgraded to moderate, low, or very low evidence.

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P7-9
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P9
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P10-11
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P9-10
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix I
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P11
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P11-12
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P12
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P12
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P11-12
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P12-13
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P12-13
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P12-13
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P12-13
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P13

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	P14
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 3
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1 & 2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Table 2
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P16-17
	23b	Discuss any limitations of the evidence included in the review.	P21
	23c	Discuss any limitations of the review processes used.	P21
	23d	Discuss implications of the results for practice, policy, and future research.	P21-22
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P9
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P9
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	No amendments
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P22
Competing interests	26	Declare any competing interests of review authors.	P22
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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